

AKADEMIKA

FOL-DUAL / PASSIVE / FIBER / CD

**ADVANCED FIBER OPTICS
TRAINING SYSTEM**

EXPERIMENTAL MANUAL

....A MESSAGE FROM**Introduction**

In today's engineering, the real challenge in the field of FIBER OPTICS is to analyze the need of the end users and to provide them with an innovative and a high-quality product. It is our vision to furnish products which are reliable and convenient to use. AKADEMIKA offers a high degree of expertise in developing efficient products.

Company's Motto

- "Empowering students to soar to great heights"

In India, we are the pioneers in the field of FIBER OPTICS SYSTEMS, committed to serve the best products. We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

AKADEMIKA LAB SOLUTIONS

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SAFETY GUIDELINES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the mains power to the modules, be sure that the switch settings and connecting chords are connected correctly, as per the experiment.
- In Advance Fiber Optic Experimental Modules, signals are to be monitored with a Digital Storage Oscilloscope, or with the help of indicators as explained in the manual. The scope input channel should be A.C. coupled, unless otherwise indicated. For observation of signals on oscilloscope use X10 (Attenuation Probe).

➤ **Laser Safety**

It is imperative that you read and understand the following safety Information. Before attempting any of the experiments from this manual. If you have any inquiries, please consult your lecturer or demonstrator.

➤ **Operational Hazard-Semiconductor Laser Diode**

The laser diode sources emit radiation at 1310nm and 1550nm, which is invisible to the human eye and may cause **eye damage** if the output beam is viewed directly. While using, never stare at the optical output port of the Laser Diode, when its dust cap is removed. Always leave the dust cap for the fiber patch cord in position until ready to make a connection and never stare at the end of a fiber patch cord connected to the Laser Diode output port or point the fiber end at yourself or anyone else in your vicinity before making a connection. Never use external optics (i.e. collimating lenses) to view the Laser Diode emission at a close range.

The output power of the semiconductor Laser Diode is less than 1mW which is in line withall the applicable standards of DHMS regulations and falls within the international safety Class3B.

- These modules must be employed only for the use for which it has been conceived, i.e. as an educational system and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer system, turn off the power supply. Please do not tamper the system. If you require our service, kindly contact the service centre for technical assistance.
- These modules are liable to malfunction/ underperform if it is not operated under following conditions:
 - Ambient temperature: Between 0 to 45 ° C
 - Relative humidity : Between 20 to 80 %
 - Avoid immediate/significant change of temperature and humidity.

WARRANTY

These modules are warranted against defects in workmanship and materials. If any failure occurs due to defect in either workmanship or material, under normal use, within a year from the original date of purchase, then such a failure will be corrected, free of charge to the purchaser by repairing or replacing the defective part or parts. But, if the failure happens as a result of the user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with some charges.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. The Type and the Model No. of the kit should be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transport expenses, required to send back all the repaired products for the moment. After repairing, the transport expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal or careless use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of the above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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INTRODUCTION

Before fiber optics came along the primary means of real time, data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances over light energy. Although, Fiber Optics was a very controversial technology initially, today, it is considered as reliable & and cost effective.

High bandwidth properties of fiber cables can be used over greater distances because of the Low Loss as compared to copper cables. In data networks, this can be as much as 2Km without the use of repeaters. Their light weight and small size also make them ideal for applications, where running copper cables would be impractical. By using multiplexers, one fiber can replace hundreds of copper cables. This is pretty impressive for a tiny glass filament, but the real benefits in the data industry are its immunity to Electro Magnetic Interference (EMI), and the fact that glass is not an electrical conductor. Fiber is non-conductive, therefore, it can be used wherever electrical isolation is required. For instance, it can be used between buildings where copper cables would require cross bonding to eliminate differences in earth potentials. Also, Fibers do not pose any threat in dangerous environments such as chemical plants, where a spark could trigger an explosion. Last but not least is the security aspect. It is very, very difficult to tap into a fiber cable to read the data signals

You will learn soon that fiber optic communication is indeed a simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today.

HOW OPTICAL FIBER WORKS?

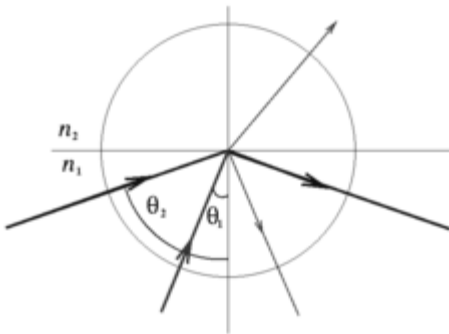
The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Think of a fiber cable in terms of a very long cardboard roll (from the inside roll of paper towel) that is coated with a mirror. If you shine a flashlight in one you can see light at the far end - even if you bend the roll around a corner.

Light pulses move easily down the fiber-optic line because of a principle known as the total internal reflection. "This principle of the total internal reflection states that when the angle of incidence exceeds a critical value, light cannot get out of the glass; instead, the light bounces back in as shown in figure. When this principle is applied to the construction of the fiber-optic strand, it is possible to transmit information through fiber lines in the form of light pulses.

Total internal reflection is an optical phenomenon that occurs when light strikes a medium boundary at a steep angle. If the refractive index is lesser on the other side of the boundary, no light can pass through, so all of the light gets reflected effectively. The critical angle is the angle of incidence above which the total internal reflection occurs.

When light crosses a boundary between materials with different refractive indices, the light beam will be partially be refracted at the boundary surface and partially it will get reflected. However, if the angle of incidence is shallower (closer to the boundary) than the critical angle (the angle of incidence where light is refracted so that it travels along the boundary),

then the light will stop crossing the boundary altogether, instead it will reflect back totally internally. This can only occur where light travels from a medium with a higher refractive index to the one with a lower refractive index. For example, it will occur when light passes from glass to air, but not when light would pass from air to glass.

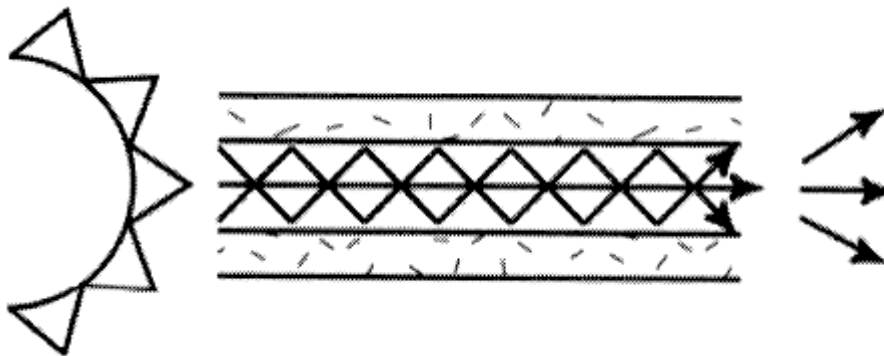


Total internal reflection

TYPES OF OPTICAL FIBERS

The three types of fiber optic cables commonly used are as follows:

- Single Mode
- Multimode
- Plastic Optical Fiber (POF)



Fiber optic cable functions as a 'light guide', guiding the light introduced at one end of the cable to the other end. The light source can either be a light-emitting diode (LED) or a laser.

The light source is pulsed on and off, and a light-sensitive receiver on the other end of the cable converts the pulses back into the digital ones and zeros, of the original signal.

Even laser light shining through a fiber optic cable is subject to loss of strength, primarily through dispersion and scattering of the light, within the cable itself. The faster the laser fluctuates, the greater the risk of dispersion. Light strengtheners, called repeaters, may be necessary to refresh the signal in certain applications.

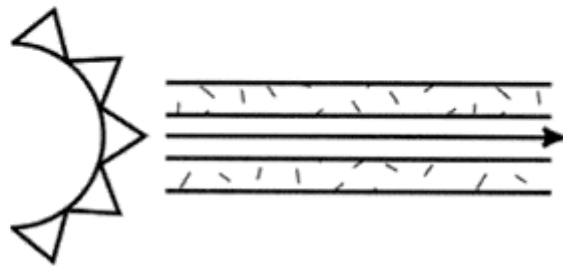
While fiber optic cable itself has become cheaper over time - an equivalent length of copper cable costs lesser per foot, but not in capacity. Fiber optic cable connectors and the equipment required to install them are more expensive than their copper counterparts.

Single Mode Cable: It is a single strand of glass fiber with a diameter of 8.3 to 10 microns that has one mode of transmission. Single Mode Fiber with a relatively narrow diameter, through which only one mode will propagate typically 1310nm or 1550nm, carries higher bandwidth than multimode fiber, but requires a light source with a narrow spectral width.

Single-mode fiber gives you a higher transmission rate and up to 50 times more distance than multimode, but it also costs more. Single-mode fiber has a much smaller core than multimode. The small core and single light-wave virtually eliminate any distortion that could result from overlapping light pulses, providing the least signal attenuation and the highest transmission speeds of any fiber cable type.

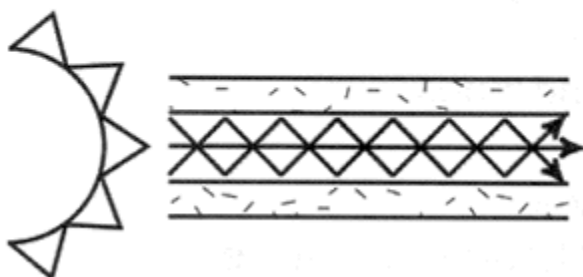
Single-mode optical fiber is an optical fiber in which only the lowest order bound mode can propagate at the wavelength of interest, typically 1300nm to 1320nm.

“Single mode fiber”
single path through the fiber



Multimode Cable: It is made of glass fibers, with common diameters in the 50-to-100 micron range for the light carry component (the most common size is 62.5 μ m). Multimode fiber gives you high bandwidth at high speeds over medium distances. Light waves are dispersed into numerous paths, or modes, as they travel through the cable's core, typically 850nm or 1300nm. Typical multimode fiber core diameters are 50, 62.5, and 100 micrometers. However, in long cable runs (greater than 3000 feet [914.4 meters]), multiple paths of light can cause signal distortion at the receiving end, resulting in an unclear and incomplete data transmission.

“Multimode fiber”
multiple paths through the fiber



Plastic Optical Fiber (POF): It is a newer plastic-based cable which promises performance, similar to glass cable on very short runs, but at a lower cost.

SINGLE MODE V/S MULTIMODE:

A fiber can support one or several (more than one) propagation modes, the intensity distributions of which are located at or immediately around the fiber core, although some of the intensity may propagate within the fiber cladding. Other modes are not restricted to the core region and are called cladding modes. The power in these are usually lost after some distance of propagation, but it can in some cases propagate over longer distances. Outside the cladding, there is typically a protective polymer coating, which gives the fiber improved mechanical strength and protection against moisture and also determines the losses for cladding modes.

Some important differences between single-mode and multimode fibers are mentioned below:

- Single-mode fibers usually have a relatively small core (with a diameter of only a few micrometers) and can guide only a single spatial mode (disregarding the fact that there are two different polarization directions), the profile of which in most cases has roughly a Gaussian shape. Changing the launch conditions only affects the launched power, while the spatial distribution of the light exiting the fiber is fixed. Efficiently launching light into a single-mode fiber usually requires a laser source with good beam quality and precise alignment of the focusing optics, in order to achieve mode matching. Also, there are large mode area fibers with single-mode guidance, where the alignment tolerances are lower in terms of position, but higher in terms of angle (which is not considered as a big problem).
- Multimode fibers usually have a larger core and / or a larger index difference between core and cladding, so that they support multiple modes with different intensity distributions (see the figure below). In this case, the spatial profile of light exiting the fiber core may depend on the launch conditions, which determine the distribution of power among the spatial modes.

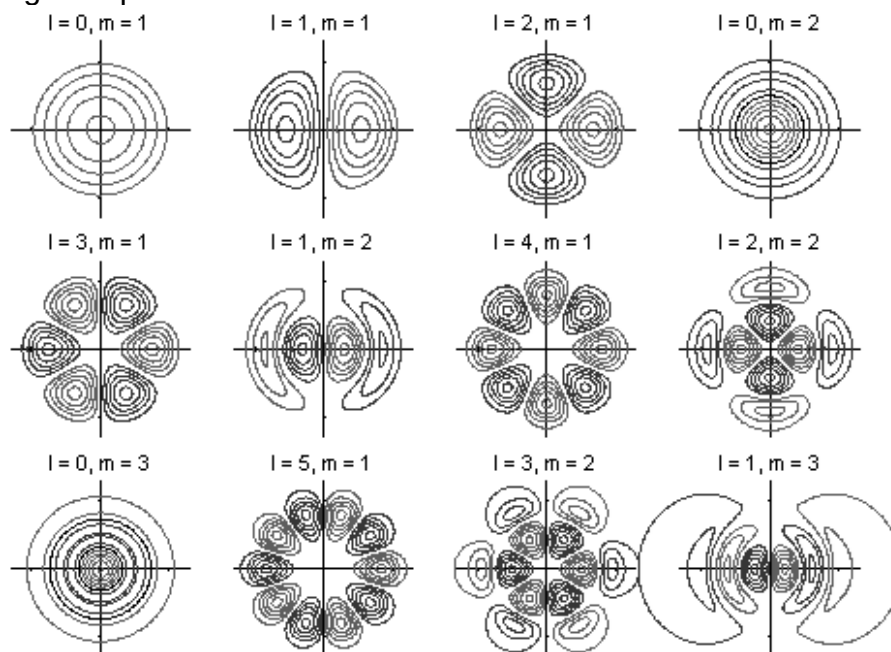


Fig.: Modes of a fiber with a top-hat refractive index profile.

The lowest-order mode ($l = 1, m = 0$, called LP_{01} mode) has an intensity profile which is similar to that of a Gaussian beam. Note that for a given wavelength, the number of fiber modes is finite; e.g., the figure shows all modes for the given configuration. In general, light launched into a multimode fiber will excite a superposition of different modes, which can have a rather complicated shape.

Long-range optical fiber communications systems usually use single-mode fibers, because the different group velocities of different modes would mess up the data at high data rates. However, for shorter distances, multimode fibers are more convenient as the demand on light sources and component alignment are lower. Therefore, local area networks (LANs), except those for highest bandwidth, normally use multimode fiber.

Single-mode fibers are also normally used for fiber lasers and amplifiers. Example: Multimode fibers are often used to transport light from a laser source to the place where it is required, particularly when the light source has a poor beam quality and / or the high optical power requires a large mode area.

Different modes of a fiber can be coupled via various effects. Example: By bending or often by irregularities in the refractive index profile. These may be unwanted or purposely introduced. E.g. As fiber Bragg gratings. Waveguides theory shows that an important factor for the coupling between different fiber modes is the difference of their wave numbers, which for efficient coupling has to match the wave number of a coupling disturbance.

ADVANTAGES OF FIBER OPTICS OVER CONVENTIONAL COPPER CABLES:

- **Less expensive** - Several miles of optical cable can be made at cheaper rates compared to equivalent lengths of copper wires. This saves your service providers (cable TV, Internet) and your money too.
- **Thinner** - Optical fibers can be drawn to smaller diameters than copper wire.
- **Higher carrying capacity** - Optical fibers are thinner than copper wires, therefore, more fibers can be bundled within a given diameter cable than the copper wires. This allows more phone lines to go over the same cable or more channels to pass through the cable into your cable TV box.
- **Less signal degradation** - The loss of signal in optical fiber is lesser than that observed in copper wires.
- **Light signals** - Unlike electrical signals in copper wires, light signals from one fiber does not interfere with those of the other fibers in the same cable. This means clearer phone conversations or TV reception.
- **Low power** - Signals in optical fibers degrade less; therefore, lower-power transmitters can be used instead of the high-voltage electrical transmitters, as required for copper wires. Again, this saves your provider and your money.
- **Digital signals** - Optical fibers are ideally suited for carrying digital information, which is especially useful in computer networks.
- **Non-flammable** - No electricity is passed through optical fibers; therefore, there is no fire hazard.
- **Lightweight** - An optical cable weighs lesser than a comparable copper wire cable. Also, Fiber-optic cables take very little space in the ground.

- **Flexible** - Fiber optics are very flexible and can transmit and receive light effectively; therefore, they are used in many flexible digital cameras for the various purposes. Some of the examples are mentioned below:
 - **Medical imaging** – Used in bronchoscopes, endoscopes and laparoscopes.
 - **Mechanical imaging** – To inspect mechanical welds along pipes and engines (airplanes, rockets, space shuttles, cars).
 - **Plumbing** - To inspect sewer lines.

With all the above properties of optical fiber, it is becoming very important in sensing fields too. Most of the properties of optical fiber are useful in sensors. Further, optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

TECHNICAL SPECIFICATIONS

A. FOL-DUAL: DUAL WAVELENGTH FIBER OPTIC LASER SOURCE AND DETECTOR

1. LASER SOURCE -1

Central wavelength	: 1310nm
Spectral width	: 2nm
Output power	: 1.8mW
Threshold current	: 12mA Max
PD monitor current	: 0.8mA

2. LASER SOURCE -2

Central wavelength	: 1550nm
Spectral width	: 1nm
Output power	: 0.9mW
Threshold current	: 10mA Max
PD monitor current	: 1.1mA

3. DETECTOR -1

Type	: PIN
Spectral Bandwidth	: 1250nm to 1600nm
Bandwidth	: 1.5GHz
Forward current	: 5mA
Reverse voltage : 30V	
Reverse current	: 500uA
Responsivity	: 0.8 A/W

4. DETECTOR -2

Type	: PIN TIA
Spectral Bandwidth	: 1150nm to 1600nm
Sensitivity	: -37dbm.
Bandwidth	: 155MHz
Data rate	: 155Mbps
Gain Differential output	: 200mV/uW
Power supply current	: 35mA
Supply voltage	: 7Vmax

5. PULSE GENERATOR

Pulse width	: Variable from 30ns to 220ns
Pulse amplitude	: 3V for 30nSec and 4V for 100nSec.

6. FIBER OPTIC CABLE

Type	: Glass fiber - single mode
Core diameter	: 62.5 & 9 micron
Clad diameter	: 125 micron
Primary coating diameter	: 250 micron
Total outer diameter	: 2.2 micron
Length of fiber	: 1 meter each

B. FOL-PASSIVE : FIBER OPTIC PASSIVE COMPONENT MODULE

1. COUPLER 2X2

IL Max	:	<= 4.10(dB)
Uniformity	:	<= 0.80 (dB)
Operating temp.	:	-40 to 85°C
Connector Type	:	ST/PC
Band Pass (nm)	:	1310 & 1550+/- 20nm
Package Type	:	MD,900u Tube

PARAMETER	O/P PORT	1310nm	1550nm
INSERTION LOSS MAX (dB)	I1-01	3.30	3.65
	I1-02	3.80	3.70
	I2-01	3.50	3.23
	I2-02	3.80	3.71
Uniformity (dB)	IN1	0.50	0.05
	IN2	0.30	0.48
OPERATING TEMPRATURE RANGE: -40 TO 85°C			

2. ISOLATOR 1310nm

Operating wavelength	:	1310nm
Insertion Loss (25°C)	:	0.33dB
Isolation (1550nm,25°C)	:	30dB
Return Loss (I/O)	:	>=60/55
Power Handling	:	<=500mW
Connector Type	:	ST/UPC

3. ISOLATOR 1550nm

Operating wavelength	:	1550nm
Insertion Loss (25°C)	:	0.36dB
Isolation (1310nm,25°C)	:	43.50dB
Return Loss (I/O)	:	>=60/55
Power Handling	:	<=500mW
Connector Type	:	ST/UPC

4. ATTENUATOR : 3dB&15dB

5. WDM 2X1

Operating wavelength	:	Port1=1310nm & Port2=1550nm
IL Max	:	<= 0.80 (dB)
Isolation Min	:	>= 15.00 (dB)
Band pass (nm)	:	1310 & 1550 (+/- 15nm)
Connector Type	:	ST/PC
Package Type	:	MD,900u Tube
TEST RESULT		

PARAMETERS	WAVELENGTHS			
	PORT	1310nm	PORT	1550nm
INSERTION LOSS (dB)	01	0.20	02	0.47
ISOLATION (dB)	02	23.81	01	19.86
DATA INCLUDES LOSS (OPERATING TEMPRATURE RANGE: - 40 TO 85°C)				

C. FOL-FIBER : SINGLE MODE FIBER OPTIC CABLE MODULE

1. FIBER-1

Length of fiber : 100 meter
Fiber : 9u/125u Single Mode Glass Fiber

2. FIBER-2

Length of fiber : 500 meter
Fiber : 9u/125u Single Mode Glass Fiber

3. FIBER-3

Length of fiber : 1000 meter
Fiber : 9u/125u Single Mode Glass Fiber

D. FOL-CD : CHROMATIC DISPERSION MODULE

1. FIBER

Length of fiber : 25 Km
Fiber : 9u/125u Single Mode Glass Fiber

E. Accessories

FOL-DUAL		Quantity
1	ST-ST Patch cord For SM Glass Fiber -1 mtr	7
2	Power cord	1
3	BNC TO BNC Cable	3
4	EXPT. Manual	1
5	BNC TO BNC 'T' Conn.	1

ABOUT ADVANCE FIBER OPTIC EXPERIMENTAL MODULES

In order to explore the advanced fiber optic techniques, FOM series of fiber optic modules are designed, which works with industrial grade fiber optic components. The FOM series of modules cover various active and passive components. By using the various combinations of modules different experiments can be performed.

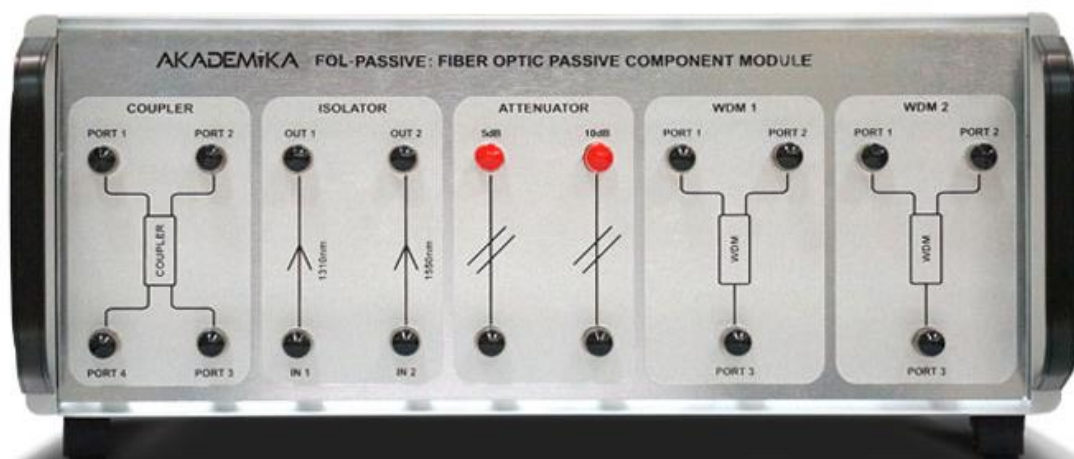
FOL-DUAL :Fiber optic source and detector module
 FOL-PASSIVE :Fiber optic passive component module
 FOL-FIBER :Multi Mode Fiber optic cable module
 FOL-CD :Chromatic Dispersion module

FOL-DUAL fiber optic source and detector module provides two numbers of laser sources at 1310nm & 1550nm with PIN detector and PIN TIA detectors. A built in pulse generator is also provided to avoid the use of external instrument. All connectors are suitable for ST type of connector interface.



FOM-DUAL: Fiber optic source and detector module

FOM-PASSIVE has all necessary passive components like coupler, isolator, attenuator and WDM.



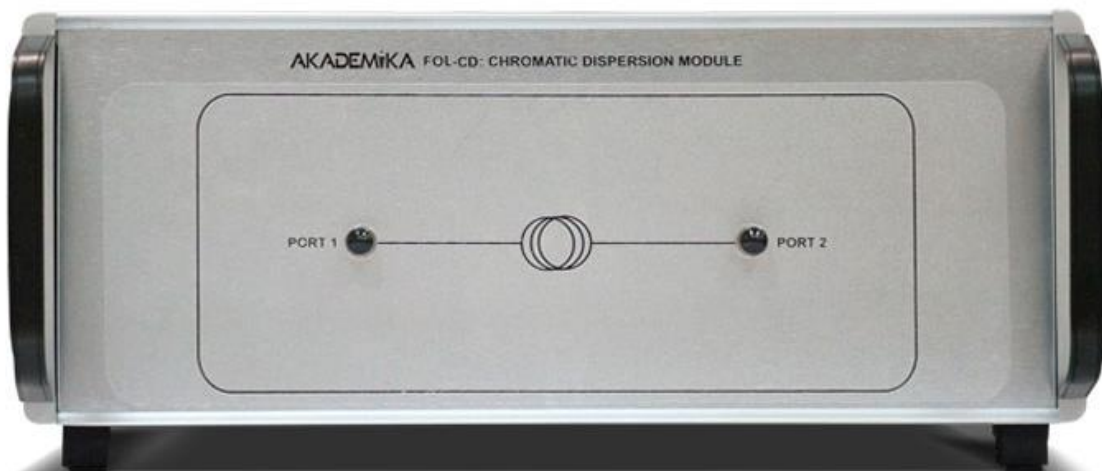
FOL-PASSIVE: Fiber optic passive component module

FOL-FIBER: Since fiber cable is made of glass and is very brittle, the students may not handle it carefully. Hence, uniform length of fibers is provided inside the cabinet and end points are provided on the front panel making it easy for the students to handle long lengths of fiber.



FOL-FIBER: Fiber optic cable module

Module FOL-CD is specially designed to perform various dispersion experiments. A special purpose fiber is provided along with WDM for the laboratory use to make the study perfect and easy.



FOL-CD : Chromatic dispersion module

EXPERIMENT NO. 1

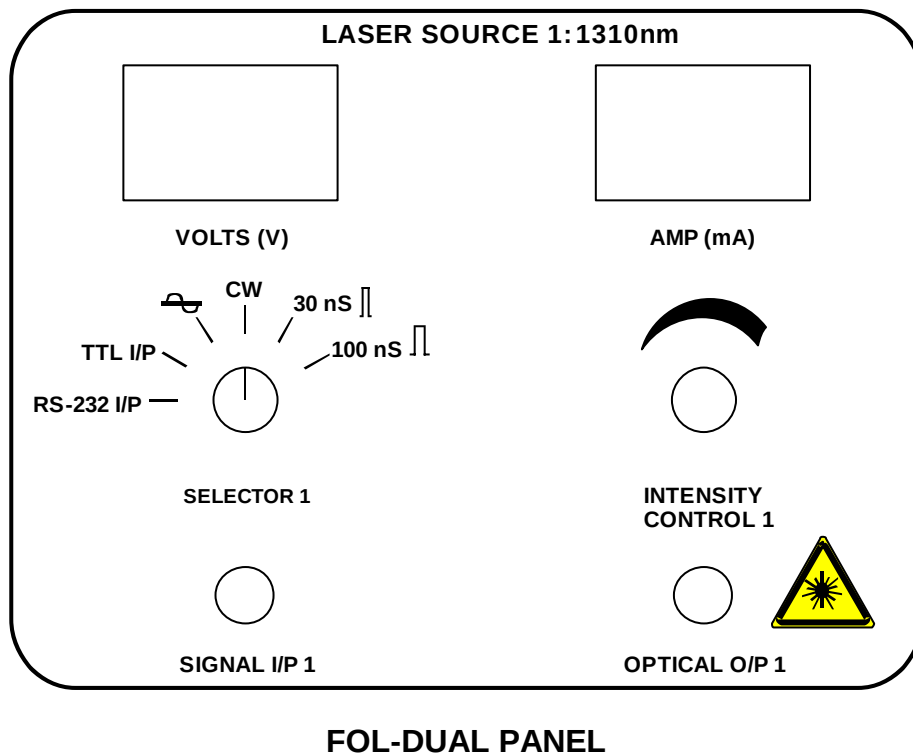


Fig.1 V-I CHARACTERISTICS OF 1310nm LASER SOURCE

EXPERIMENT NO. 1A

OBJECTIVE

To study and plot the V-I characteristics for 1310nm Laser source for the CW

EQUIPMENTS

- FOL-DUAL module
- ST to ST Single mode 1mtr fiber optic Patch cords

PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector 1 Switch of 1310nm laser on CW position.
4. Turn ON the FOL-DUAL module.
5. Turn the Intensity Control pot slightly in the clockwise direction, until the Current display shows 00.1mA.
6. Record the current in mA and the corresponding voltage in volts.
7. Increase the current flowing through the Laser Source 1 by moving the Intensity1 pot in clockwise direction.
8. Repeat the procedure for current readings in steps of 00.1mA up to the 45mA. Record the current in mA and the corresponding voltage in volts.
9. Plot the graph for Current (mA) v/s Voltage (v).
10. Find out the Threshold Current (I_{th}) in mA from the I-V graph.

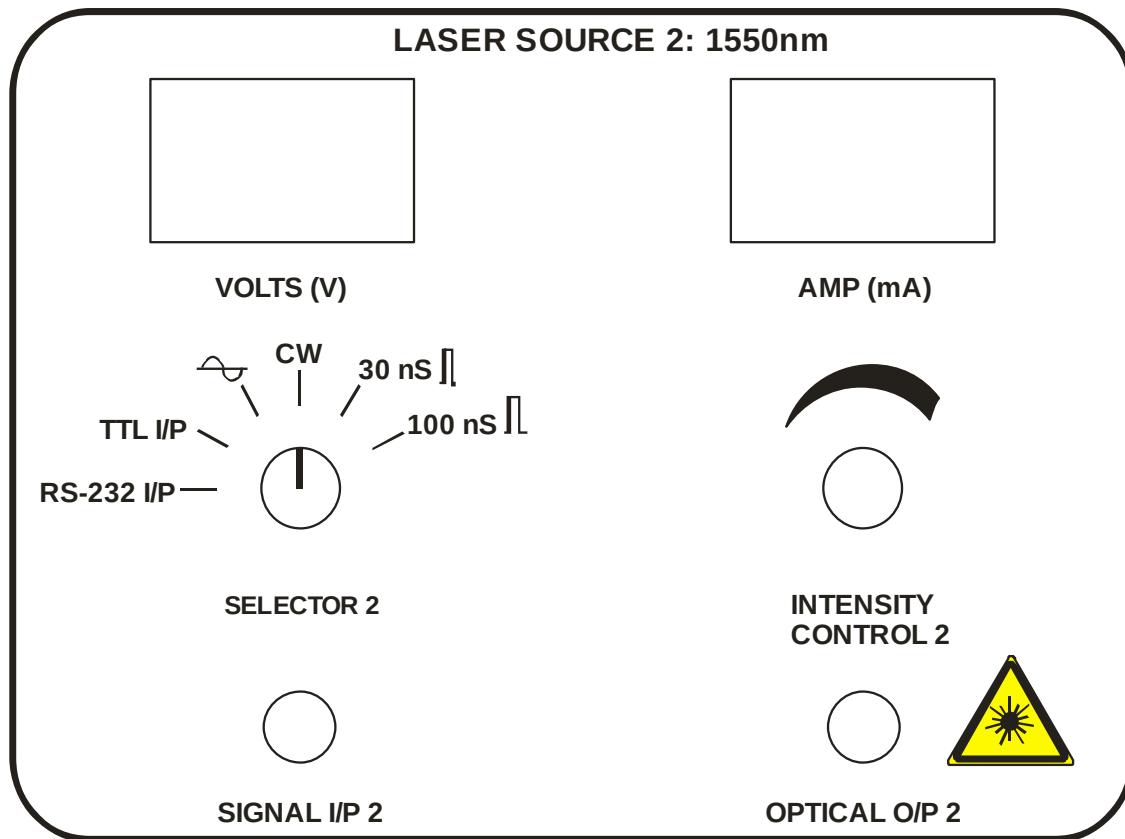
OBSERVATION TABLE

I-V characteristics of laser diode is a graph of forward voltage (V_f) across laser diode and forward current (I_f) through laser diode.

Sr. No.	FORWARD VOLTAGE (V_f)...V	FORWARD CURRENT (I_f)...mA
1	0.1	
2	0.2	
3	0.3	
4	0.4	
5	0.5	
6	0.6	
7	0.7	
8	0.8	
9	0.9	
10	1.0	
11	1.1	
12		

CONCLUSION

From the I-V graph plotted for the 1310nm Laser Source, we study the behavior of laser and measure the I_{th} (threshold current)



FOL-DUAL PANEL

Fig.1 V-I CHARACTERISTICS OF 1550nm LASER SOURCE

EXPERIMENT NO.1B

OBJECTIVE

To study and plot the V-I characteristics for 1550nm Laser source for the CW

EQUIPMENTS

- FOL-DUAL module
- ST to ST Single mode 1mtr fiber optic Patchcords

PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector2 Switch of 1550nm on CW position.
4. Turn ON the FOL-DUAL module.
5. Turn the Intensity Control pot slightly in the clockwise direction, until the Current display shows 00.1mA.
6. Record the current in mA and the corresponding voltage in volts.
7. Increase the current flowing through the Laser Source 1 by moving the Intensity 1 pot in clockwise direction.
8. Repeat the procedure for current readings in steps of 00.1mA up to the 45mA. Record the current in mA and the corresponding voltage in volts.
9. Plot the graph for Current (mA) vs Voltage (v).
10. Find out the Threshold Current (I_{th}) in mA from the I-V graph.

OBSERVATION TABLE

I-V characteristics of laser diode is a graph of forward voltage (V_f) across laser diode and forward current (I_f) through laser diode.

Sr. No.	FORWARD VOLTAGE (V_f)...V	FORWARD CURRENT (I_f)...mA
1	0.1	
2	0.2	
3	0.3	
4	0.4	
5	0.5	
6	0.6	
7	0.7	
8	0.8	
9	0.9	
10	1.0	
11	1.1	
12		

CONCLUSION

From the I-V graph plotted for the 1550 nm Laser Source, we study the behavior of laser and measure the I_{th} (threshold current)

EXPERIMENT NO. 2

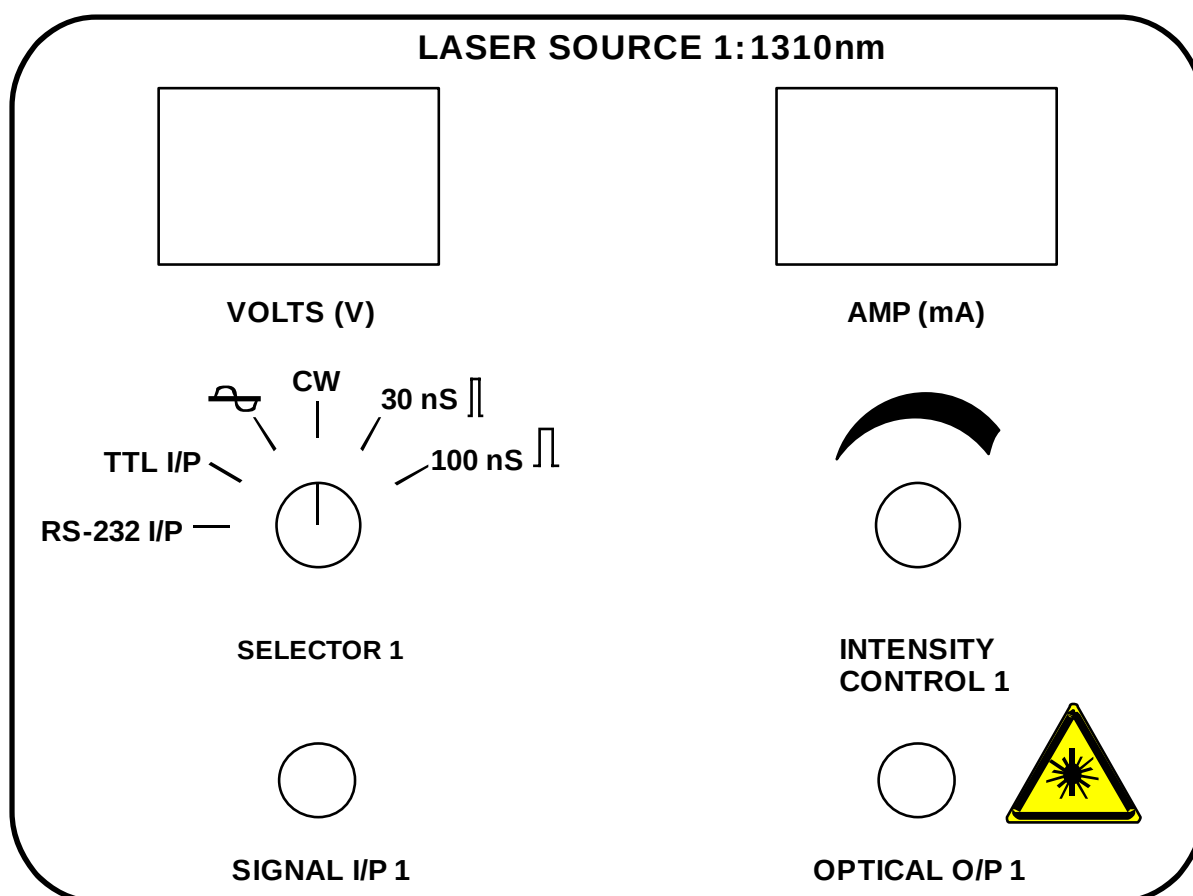


FIG-1 P-I characteristics for 1310nm Laser source

EXPERIMENT NO.2A

OBJECTIVE

To study and plot the P-I characteristics for 1310nm Laser Source for the CW

EQUIPMENTS

- FOL-DUAL module
- Optical Power Meter
- Optical Patch cords

PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector 1 Switch of 1310nm on CW position.
4. Connect Optical O/P 1 to Optical power meter, using 1meter ST-ST fiber optic cable.
5. Turn ON the FOL-DUAL module.
6. Turn the Intensity Control 1 pot slightly in clockwise direction, until the Current display shows 00.1mA.
7. Record the current in mA and the corresponding detected Optical Power (dB) in optical power meter.
8. Repeat the procedure for current readings in steps of 1mA upto 45mA. Record the current in mA and the corresponding detected Optical Power (dB) in optical power meter.
9. Plot the graph for Current (mA) vs Detected Optical Power (dB).
10. Find out the Threshold Current (I_{th}) in mA from the P-I graph.

OBSERVATION TABLE

Sr. No.	FORWARD CURRENT (I_f)...mA	OPTICAL POWER (P)...dB
1	5	
2	10	
3	15	
4	20	
5	25	
6	30	
7	35	
8	40	
9	45	

CONCLUSION

From the I-V graph plotted for the 1310nm Laser Source, Threshold Current (I_{th}) = mA.

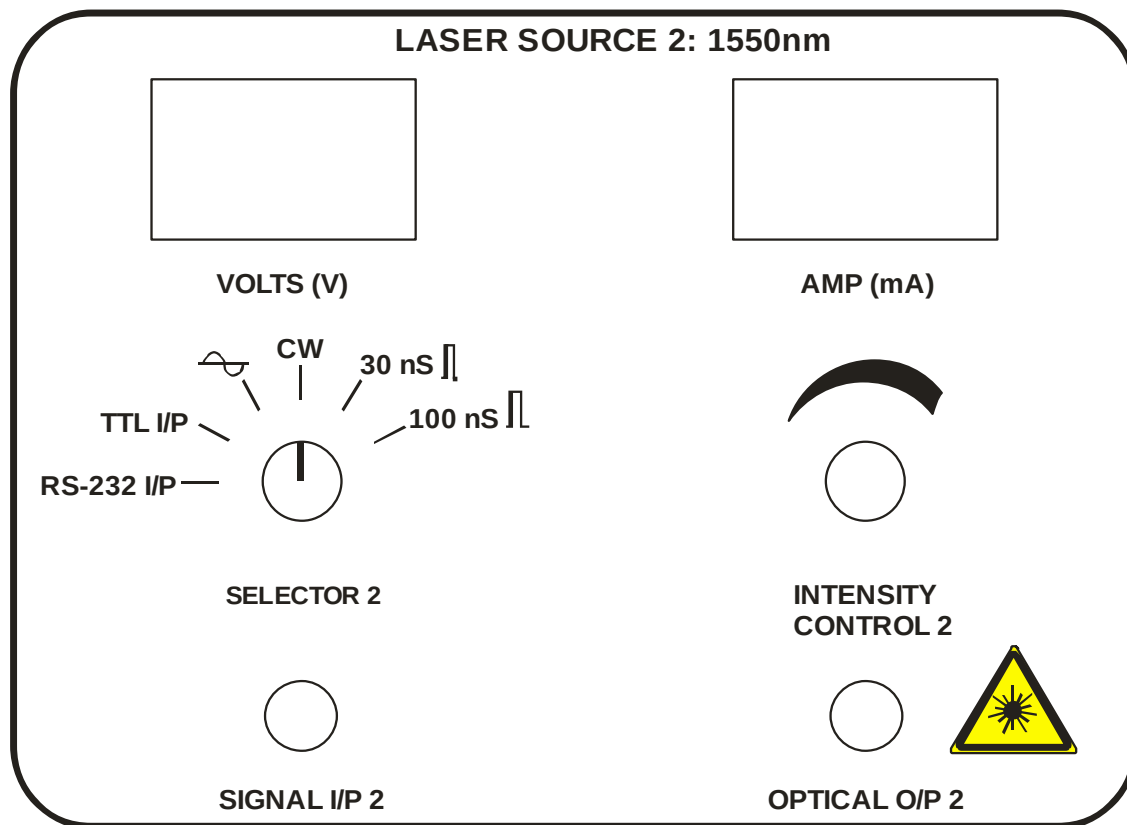


Fig.2: P-I characteristics for 1550nm Laser source

EXPERIMENT NO. 2B

OBJECTIVE

To study and plot the P-I characteristics for 1550nm Laser Source for the CW

EQUIPMENTS

- FOL-DUAL module
- Optical Power Meter
- ST to ST Single mode 1mtr fiber optic Patch Cords

PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector 2 Switch_1550nm on CW position.
4. Connect Optical O/P 2 to Optical power meter, using 1meter ST-ST fiber optic cable.
5. Turn ON the FOL-DUAL module.
6. Turn the Intensity Control 2 pot slightly in the clockwise direction, till the Current display shows 00.1mA.
7. Record the current in mA and the corresponding detected optical power (dB) in power meter.
8. Repeat the procedure for current readings in steps of 00.1mA upto 45mA. Record the current in mA and the corresponding detected optical power (dB) in the optical power meter.
9. Plot the graph for Current (mA) vs. detected optical power (dB).
10. Find out the Threshold Current (I_{th}) in mA from the I-P graph.

OBSERVATION TABLE

Sr. No.	FORWARD CURRENT (I_f)...mA	OPTICAL POWER (P)...dB
1	5	
2	10	
3	15	
4	20	
5	25	
6	30	
7	35	
8	40	
9	45	

CONCLUSION

From the I-P graph plotted for the 1550nm Laser Source,
Threshold Current (I_{th}) = mA.

**EXPERIMENT
NO. 3**

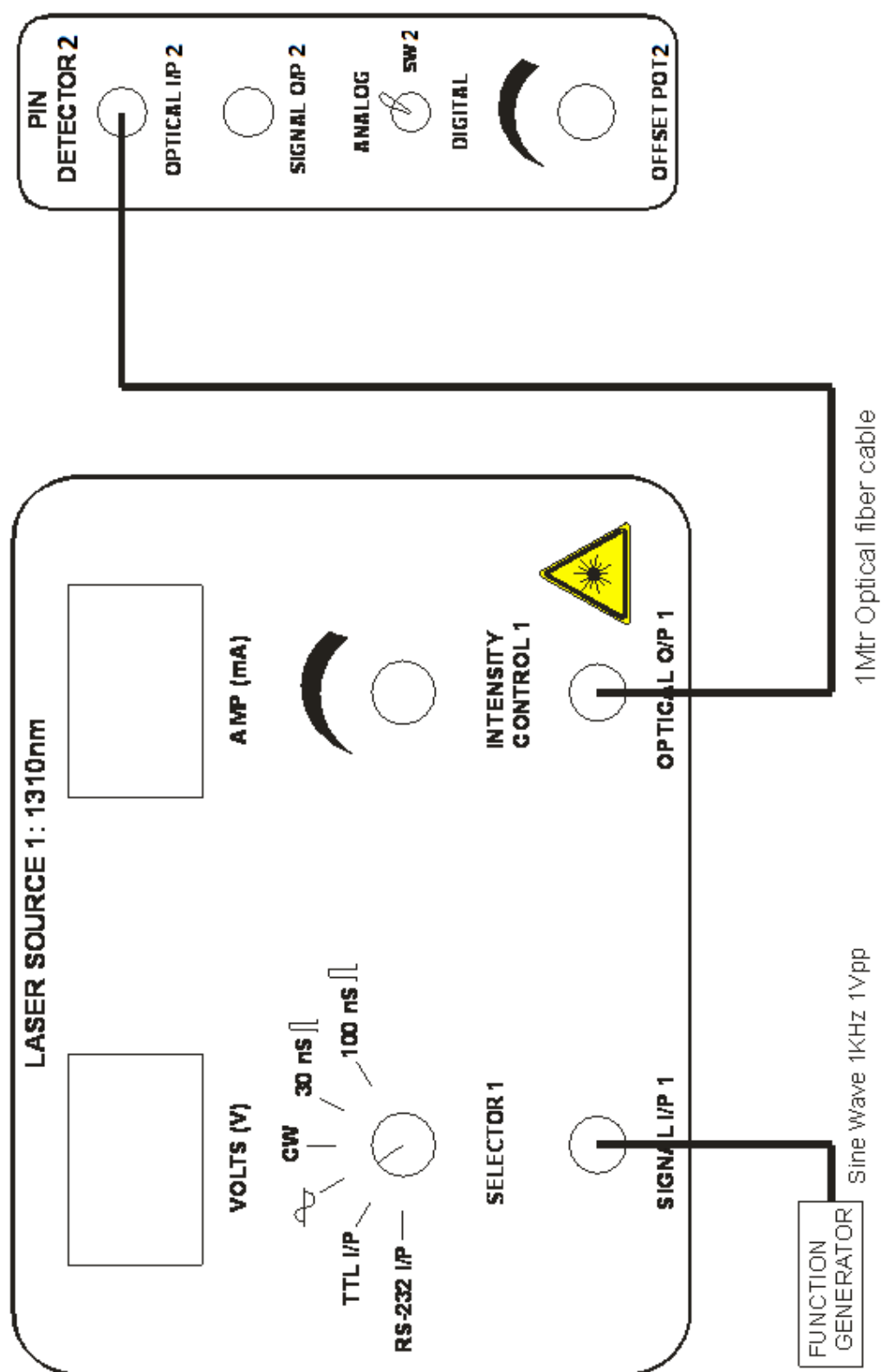


FIG.1 ANALOG COMMUNICATION USING 1310nm LASER SOURCE

EXPERIMENT NO. 3A

OBJECTIVE

To study the Analog Signal transmission for 1310nm Laser source.

EQUIPMENTS

- FOL-DUAL module
- Function Generator 2MHz
- Digital Storage Oscilloscope
- ST to ST Single mode 1mtr fiber optic Patch cords
- BNC-BNC cable

INTRODUCTION

FOL-DUAL designed to conduct experiment from BASIC (simple analog, digital links) to ADVANCE level (chromatic dispersion & OTDR)

Chromatic dispersion & OTDR required transmission of short pulse of source with sufficient power, because of which the average power during continuous analog signal such as sine wave is comparatively high.

While performing the experiments continuous analog modulations as well as pulse modulation make use of the same analog driver circuits.

Detector 1 is design with PIN PHOTO DIODE with external transimpedance with sufficient high gain.

When performing experiment such as forming simple analog link, the transmitted power is high enough to put receiver in to saturation because of we observed distortion in the received signal, to avoid this it is recommended to attenuate the transmitter power using 5dB / 10dB attenuator while performing the experiment.

PROCEDURE

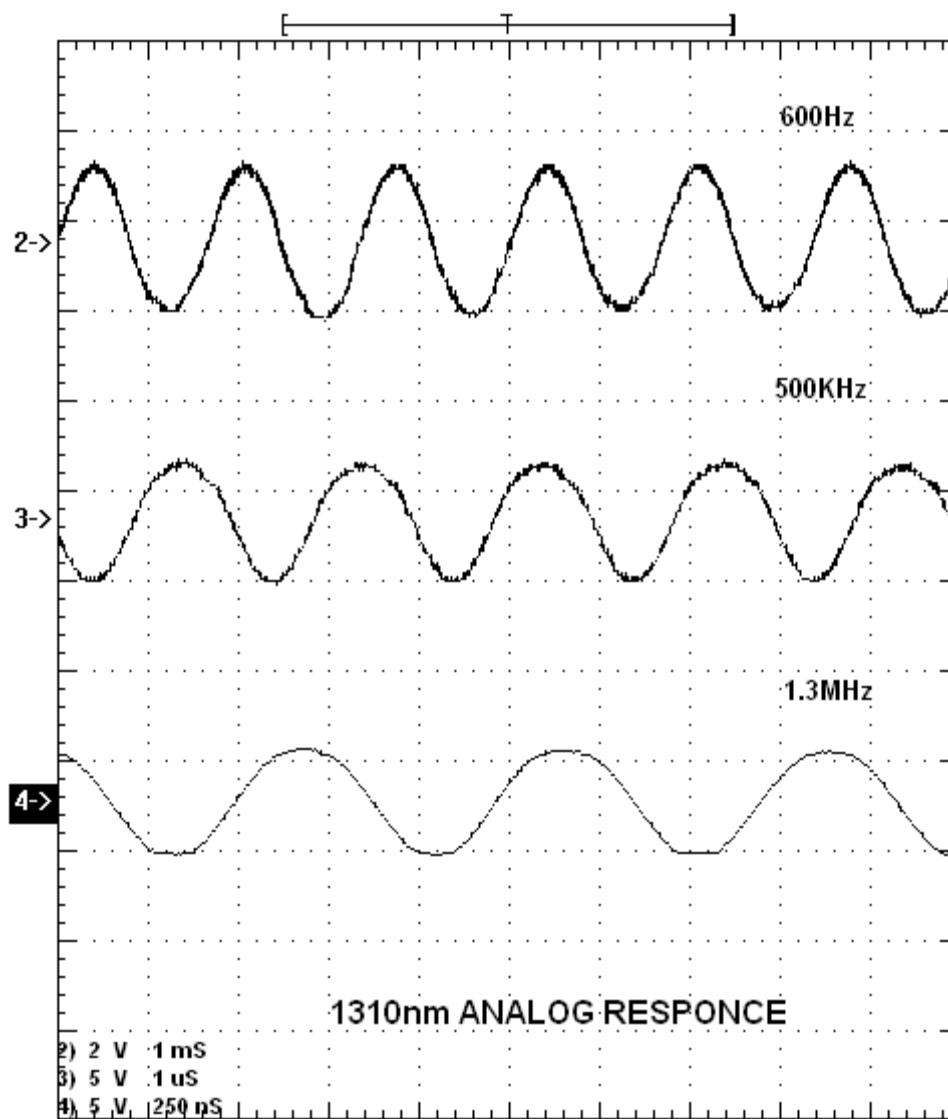
Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector1 Switch of 1310nm on analog signal (~) position and SW2 switch of PIN DETECTOR 2 on Analog mode.
4. Connect OPTICAL O/P 1 of 1310nm to Optical I/P 2 of PIN DETECTOR 2, using 1meter ST-ST fiber optic cable.
5. Set the function generator output to Sine wave_15 KHz,1Vpp and feed this signal to SIGNAL I/P 1 of 1310nm Laser on FOL-DUAL
6. Turn ON the FOL-DUAL module.
7. Observe the detected signal on SIGNAL O/p 2. Use OFFSET POT 2 to adjust the output.
8. Go on increasing the Function Generator sine wave O/p frequency in steps of 1 KHz and observe the detected signal.

CONCLUSION

Thus, we study the analog transmission of signal through fiber optic cable.

Analog Response of 1310nm_laser board_DET-2



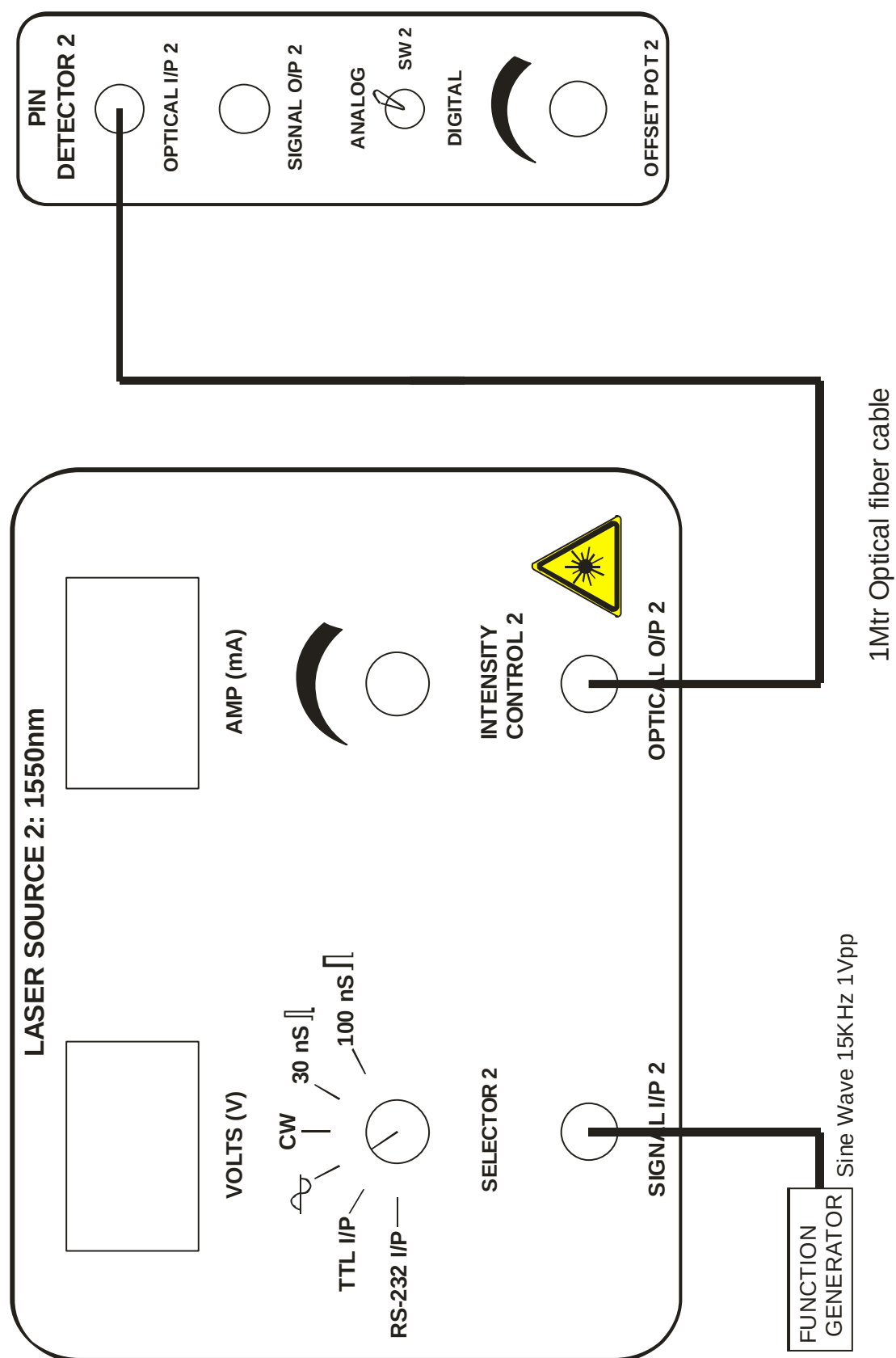


FIG.1 ANALOG COMMUNICATION USING 1550nm LASER SOURCE

EXPERIMENT NO. 3B

OBJECTIVE

To study the Analog Signal transmission for 1550nm Laser source

EQUIPMENTS

- FOL-DUAL module
- Function Generator 2MHz
- Digital Storage Oscilloscope
- ST to ST Single mode 1mtr fiber optic Patch cords
- BNC-BNC cable

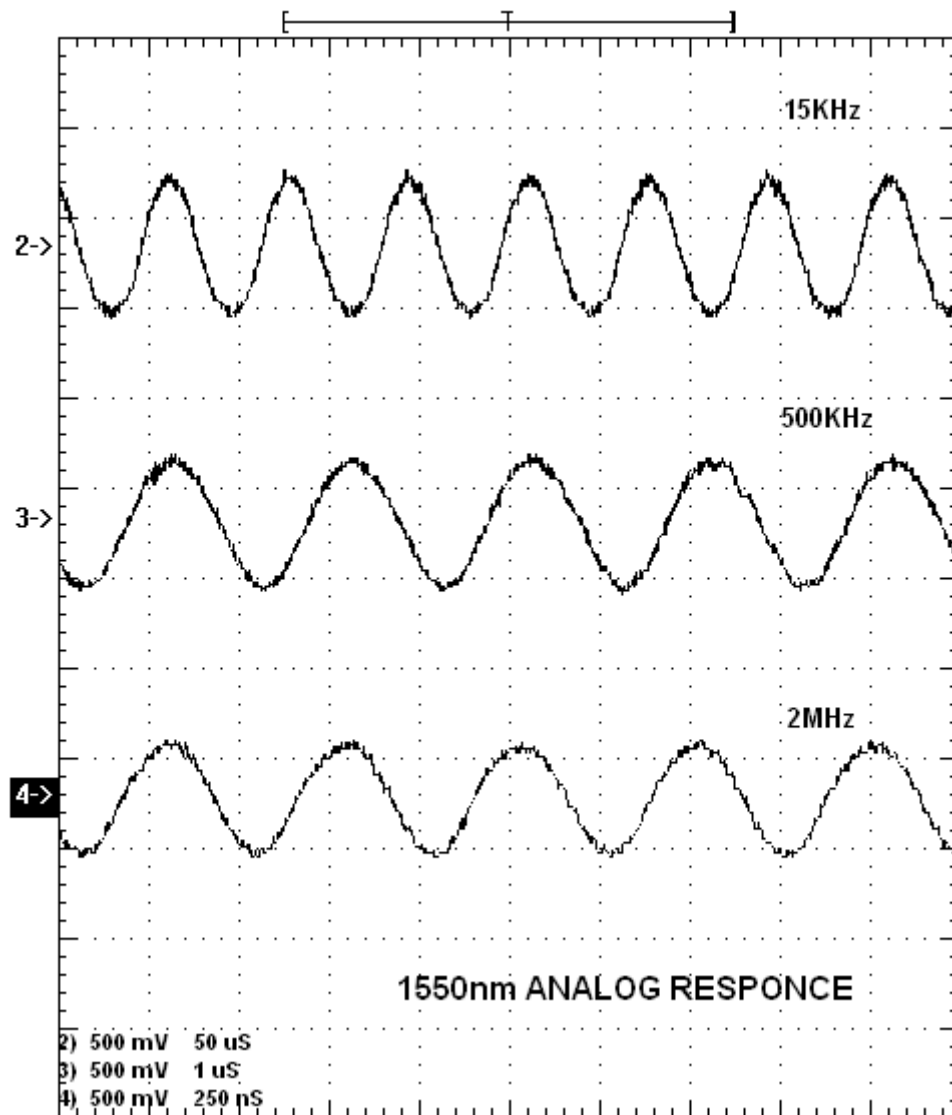
PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector 2 Switch of 1150nm on analog signal (~) position and SW2 switch of PIN DETECTOR 2 on Analog mode.
4. Connect OPTICAL O/P 1 of 1550nm to Optical I/P 2 of PIN DETECTOR 2, using 1meter ST-ST fiber optic cable.
5. Set the function generator output to Sine wave_15 KHz,1Vpp and feed this signal to SIGNAL I/P 2 of 1550nm Laser on FOL-DUAL
6. Turn ON the FOL-DUAL module.
7. Observe the detected signal on SIGNAL O/p 2. Use OFFSET POT 2 to adjust the output.
8. Go on increasing the Function Generator sine wave O/p frequency in steps of 1 KHz and observe the detected signal.

CONCLUSION

Thus, we study the analog transmission of signal through fiber optic cable.

Analog Response of 1550nm_laser board_DET-2

EXPERIMENT NO. 4

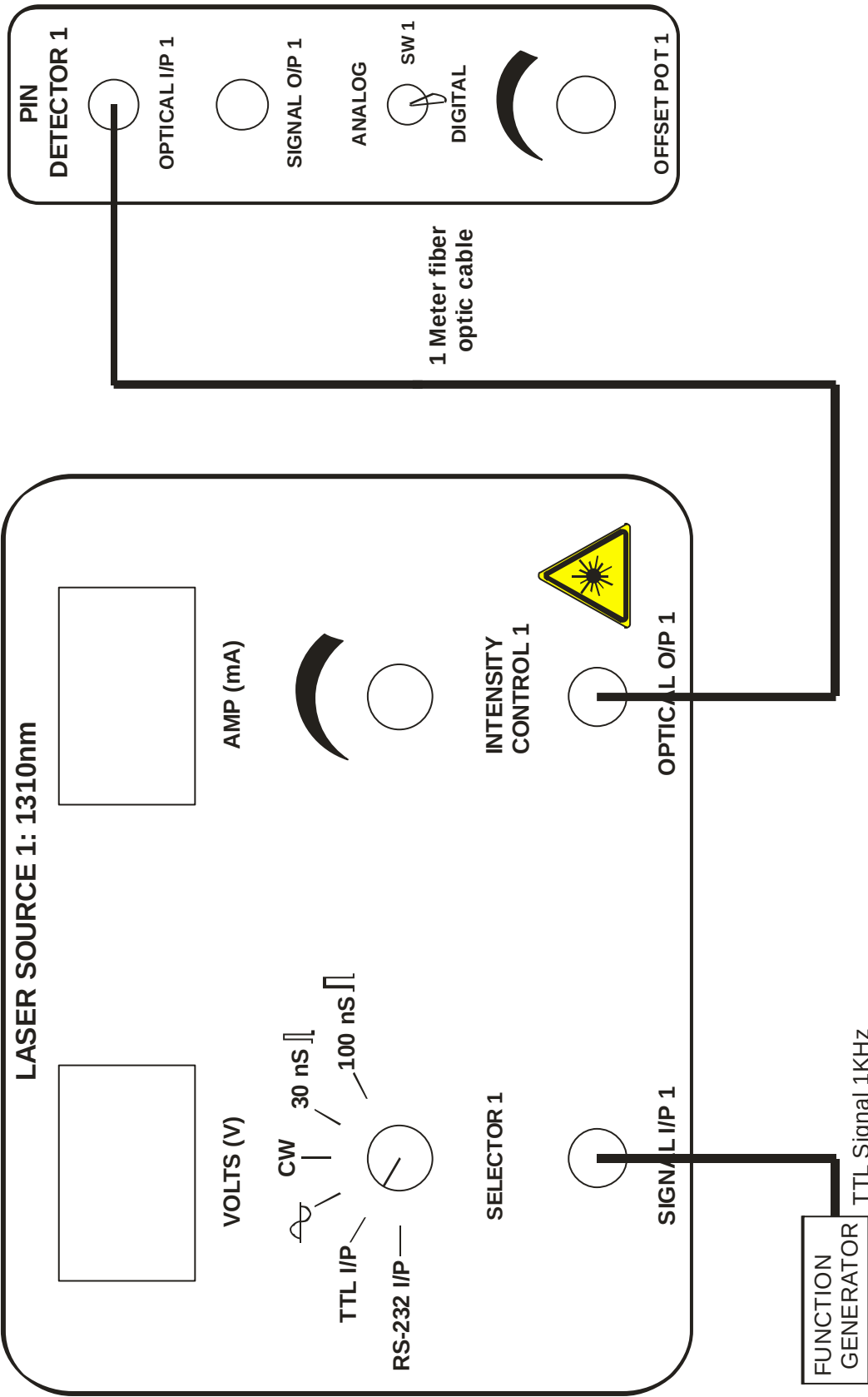


FIG.1 DIGITAL COMMUNICATION USING 1310nm LASER SOURCE

EXPERIMENT NO. 4A

OBJECTIVE

To study the Digital Signal transmission for 1310nm Laser source for the TTL signal input.

EQUIPMENTS

- FOL-DUAL module
- ST to ST Single mode 1mtr fiber optic Patch cords
- BNC-BNC cable
- Digital Oscilloscope

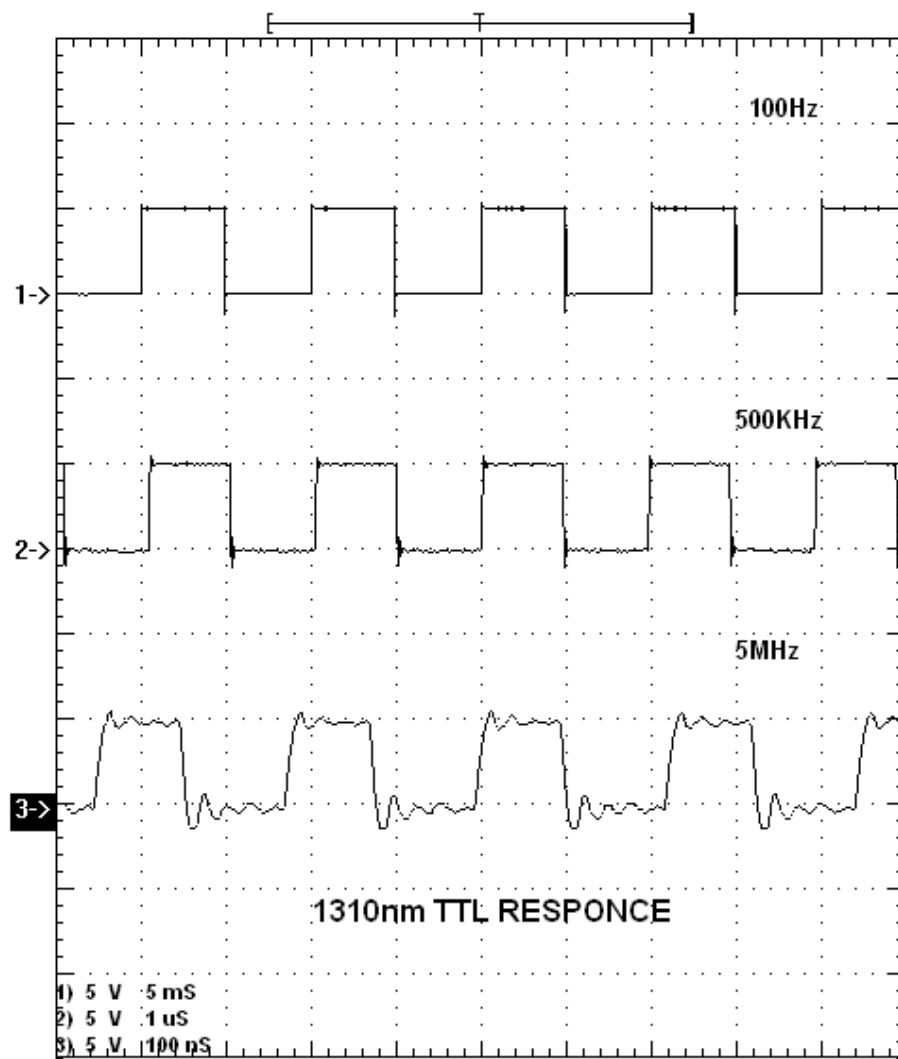
PROCEDURE

Referring to the Block Diagram,

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector Switch_1310nm on TTL position and SW1 switch of PIN DETECTOR 1 on Digital mode.
4. Connect Optical O/P 1 to Optical I/P 1, using 1meter ST-ST fiber optic cable.
5. Set the function generator output to TTL_1 KHz and feed this signal to SIGNAL I/P 1 of 1310nm Laser on FOL-DUAL
6. Turn ON the FOL-DUAL module.
7. Turn the Intensity Control pot in clockwise direction, until the Current display shows 20mA.
8. Observe the detected signal on SIGNAL O/p 1. Use OFFSET POT 1 to adjust the output.
9. Go on increasing the Function Generator TTL O/p frequency in steps of 1KHz and observe the Detected Signal o/p 1.

CONCLUSION

Thus, we study the Digital transmission of signal through fiber optic cable.

TTL Response of 1310nm_laser board_DET-1

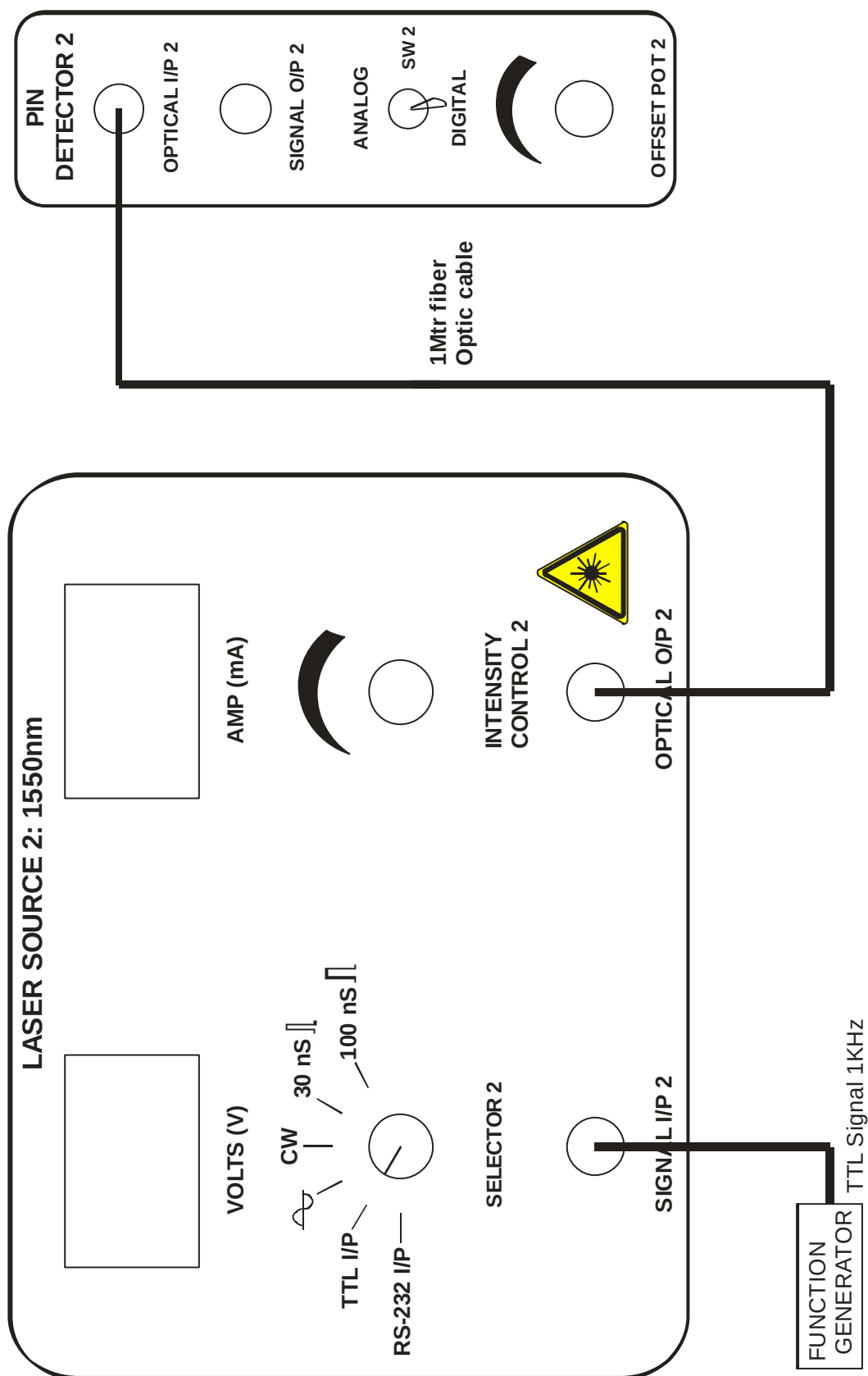


FIG.2 DIGITAL COMMUNICATION USING 1550nm LASER SOURCE

EXPERIMENT NO. 4B

OBJECTIVE

To study the Digital Signal transmission for 1550nm Laser source for the TTL signal input.

EQUIPMENTS

- FOL-DUAL module
- ST to ST single mode 1mtr fiber optic Patch cords
- BNC-BNC cable
- Digital Oscilloscope.

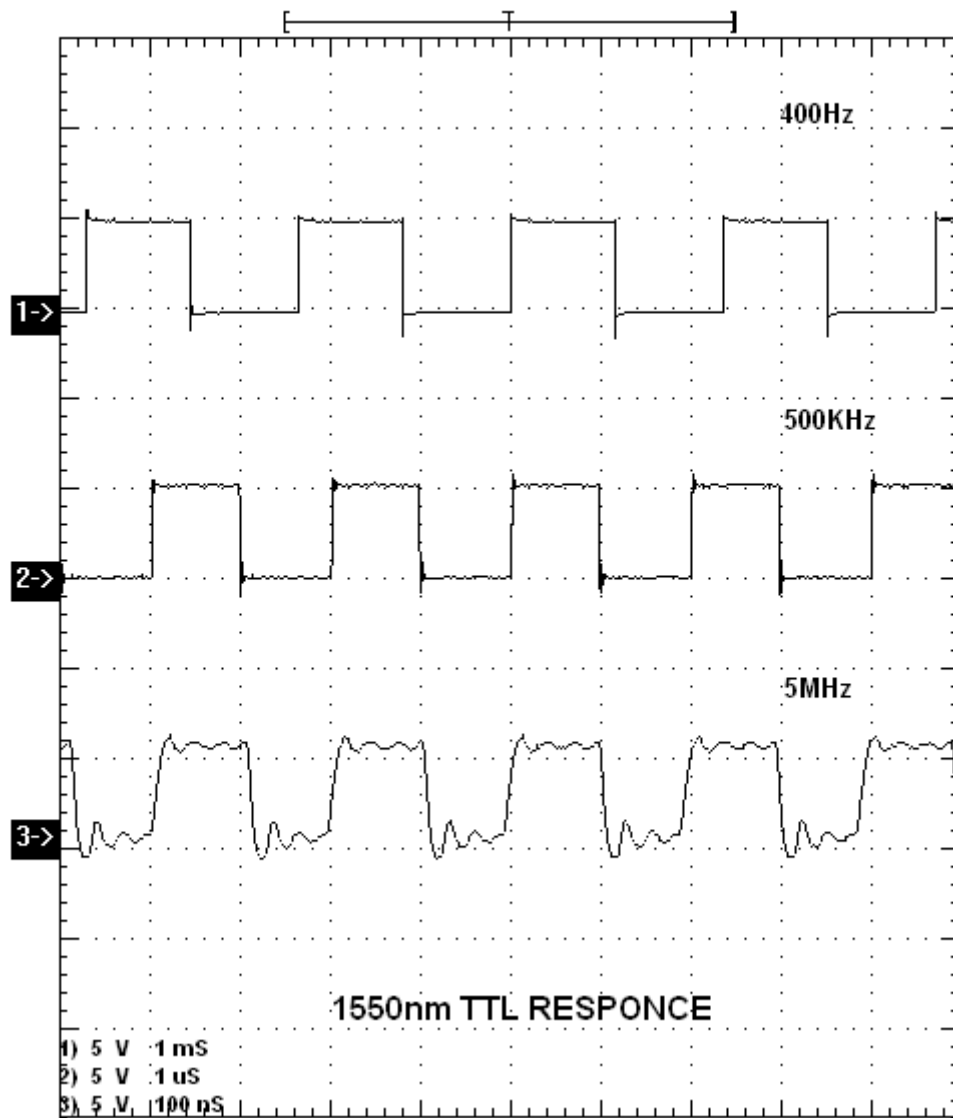
PROCEDURE

Referring to the Block Diagram,

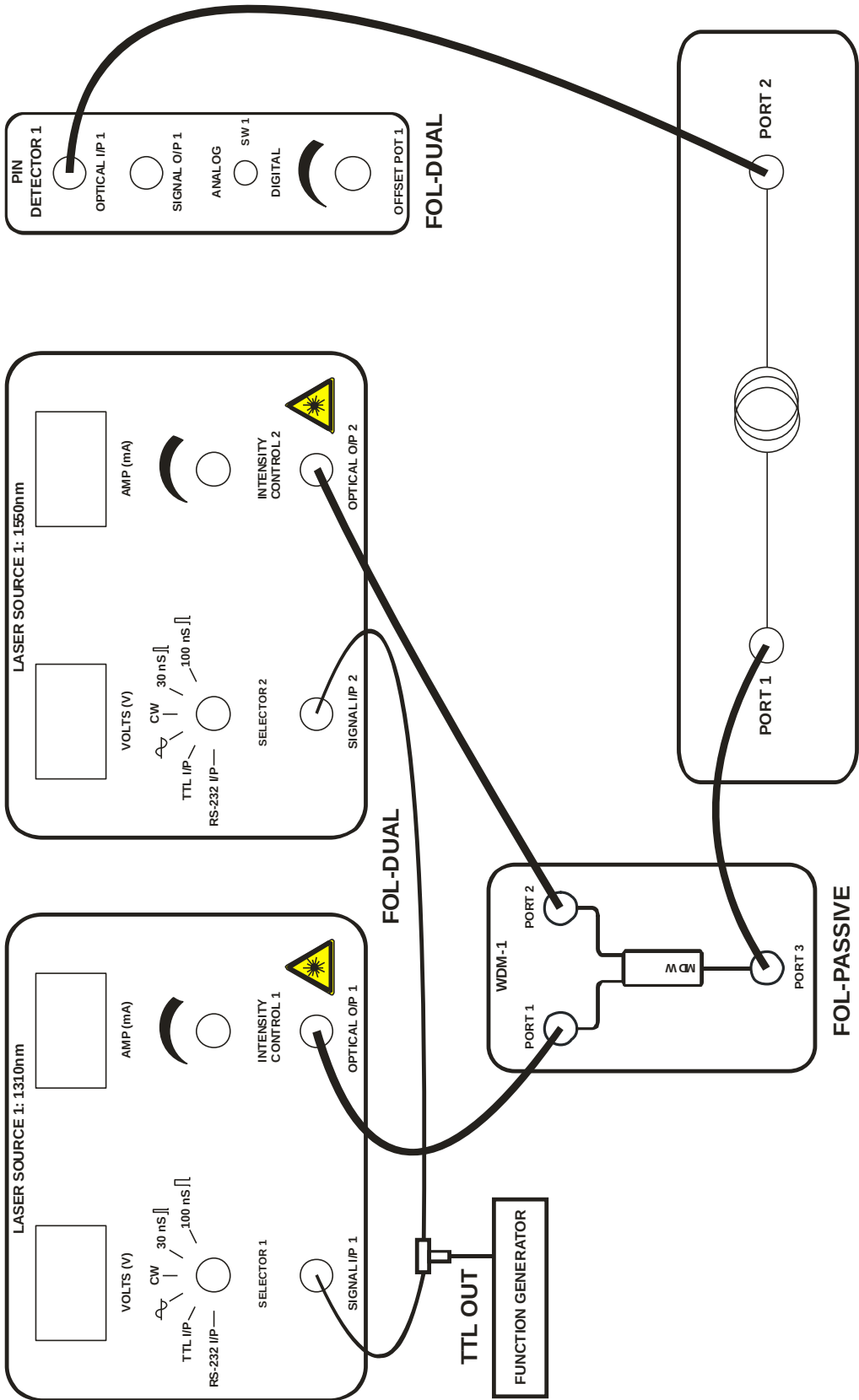
1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Keep Signal Selector Switch_1550nm on TTL position and SW2 switch of PIN DETECTOR 2 on Digital mode.
4. Connect Optical O/P 2 to Optical I/P 2, using 1meter ST-ST fiber optic cable.
5. Set the function generator output to TTL_1 KHz and feed this signal to SIGNAL I/P 2 of 1550nm Laser on FOL-DUAL
6. Turn ON the FOL-DUAL module.
7. Turn the Intensity Control pot in clockwise direction until the Current display shows 20mA.
8. Observe the detected signal on Signal O/p 2. Use OFFSET POT 2 to adjust the output.
9. Go on increasing the Function Generator TTL O/p frequency in steps of 1 KHz and observe the Detected SIGNAL O/p 2.

CONCLUSION

Thus, we study the Digital transmission of signal through fiber optic cable.

TTL Response of 1550nm_laser board_DET-2

EXPERIMENT NO. 5



Block diagram for Chromatic Dispersion

EXPERIMENT NO. 5

OBJECTIVE

To study Chromatic dispersion using dual source of 1310nm & 1550nm laser diodes.

EQUIPMENTS

- FOL-DUAL module
- FOL-PASSIVE module (Need to be purchased separately)
- FOL-CD Need to be purchased separately)
- Function Generator
- Digital Storage Oscilloscope
- Optical Patch chords
- BNC-BNC cable

THEORY

The speed of propagation of a pulse is the group velocity V_g . By definition, the group time is the opposite of the group velocity: $\tau_g = 1/V_g$. In a monomode fiber τ_g depends on the wavelength as the medium is dispersive. By definition, the dispersion D of a fiber is the derivative of group time in comparison to the wavelength:

$$D = d\tau_g / d\lambda$$

In general, D is expressed in ps/km/nm. For a standard fiber, the variation of D in relation to the wavelength is represented in figure 1.

That is, two separated pulses in a wavelength of $\Delta\lambda$ around a central wavelength λ_0 which are sent, simultaneously, into a monomode fiber. If $\Delta\lambda$ is relatively weak, D varies little in relation to λ and the time gap separating the two pulses at the end of a distance L of Fiber will be L :

$$\Delta\tau = D_0 L \Delta\lambda$$

Where D_0 is the value of D on the interval $\Delta\lambda$. The relation (1) remains valid if D varies linearly with; in this case D_0 is the value of D in λ_0 .

In this experiment we will assume that D varies linearly with λ between 1300nm and 1550nm.

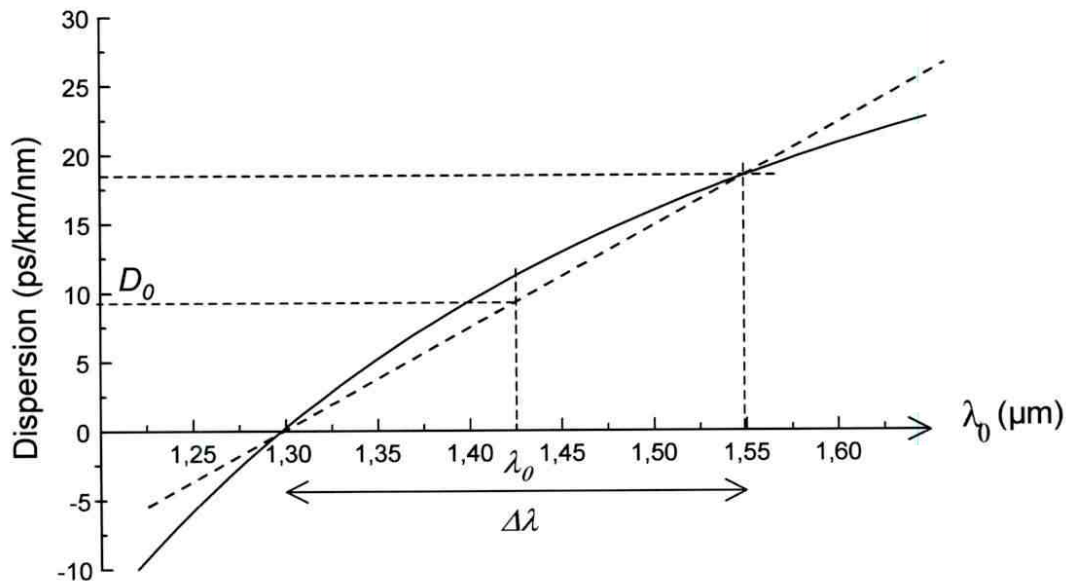


Figure 1 : Dispersion in a monomode fibre

PROCEDURE

1. Make sure that the FOL-DUAL module is off.
2. Keep Intensity Control pot fully anticlockwise.
3. Connect the Function Generator output to SIGNAL I/P 1 of 1310nm and SIGNAL I/P2 of 1550nm using BNC T connector supplied along with this set.
4. Set the Function Generator TTL o/p at 1 KHz frequency.
5. Keep Selector switch SW1 & SW2 on 30nS pulse.
6. Connect OPTICAL O/P 1 of 1310nm to Port-1 (1310nm port) of a WDM-1 of FOL-PASSIVE.
7. Connect OPTICAL O/P 2 to Port-2 (1550nm port) of a WDM-1 in FOL-PASSIVE.
8. Now connect the multiplexed Optical Signal at Port-3 of WDM 1 in FOL-PASSIVE to Port-1 of FOL-CD. (25 Km Single Mode Fiber Optic Cable).
9. Connect Port-2 of FOL-CD module to the OPTICAL I/P2 on FOL-DUAL module.
10. Connect Signal O/p 2 to Digital Oscilloscope.
11. Turn On the FOL-DUAL module.
12. Now we have passed multiplexed 30ns pulse at 1310nm and 1550nm through 25 Km spool of fiber. The spreading of two signals will take place and will be seen at the end of 25 Km fiber. Two prominent peaks that will be observed correspond to individual wavelength. To measure the Δt make use of timer cursor of the oscilloscope and measure the Δt .

Note: In case if you observe that the amplitude of chromatic dispersion waveform is very less (say around 1v or so), use 5dB attenuator in FOL-PASSIVE to attenuate power of 1550nm source output before connecting it to WDM.

CALCULATIONS

Source 1= 1310nm

Source 2= 1550nm

$$\Delta t = D_0 L \Delta \lambda$$

Δt = difference between group time pulse at 1310nm and 1550nm

D_0 = is the value of D on the interval $\Delta \lambda$. The relation is valid if D varies linearly with λ , in this case D_0 is the value of D at λ_0 .

Now as per the experimental setup we observe that

$$\Delta t = 44 \text{ ns} = 44 \times 10^3 \text{ ps}$$

$$L = 25 \text{ Km}$$

$$\Delta \lambda = 1550 \text{ nm} - 1310 \text{ nm} = 240 \text{ nm}$$

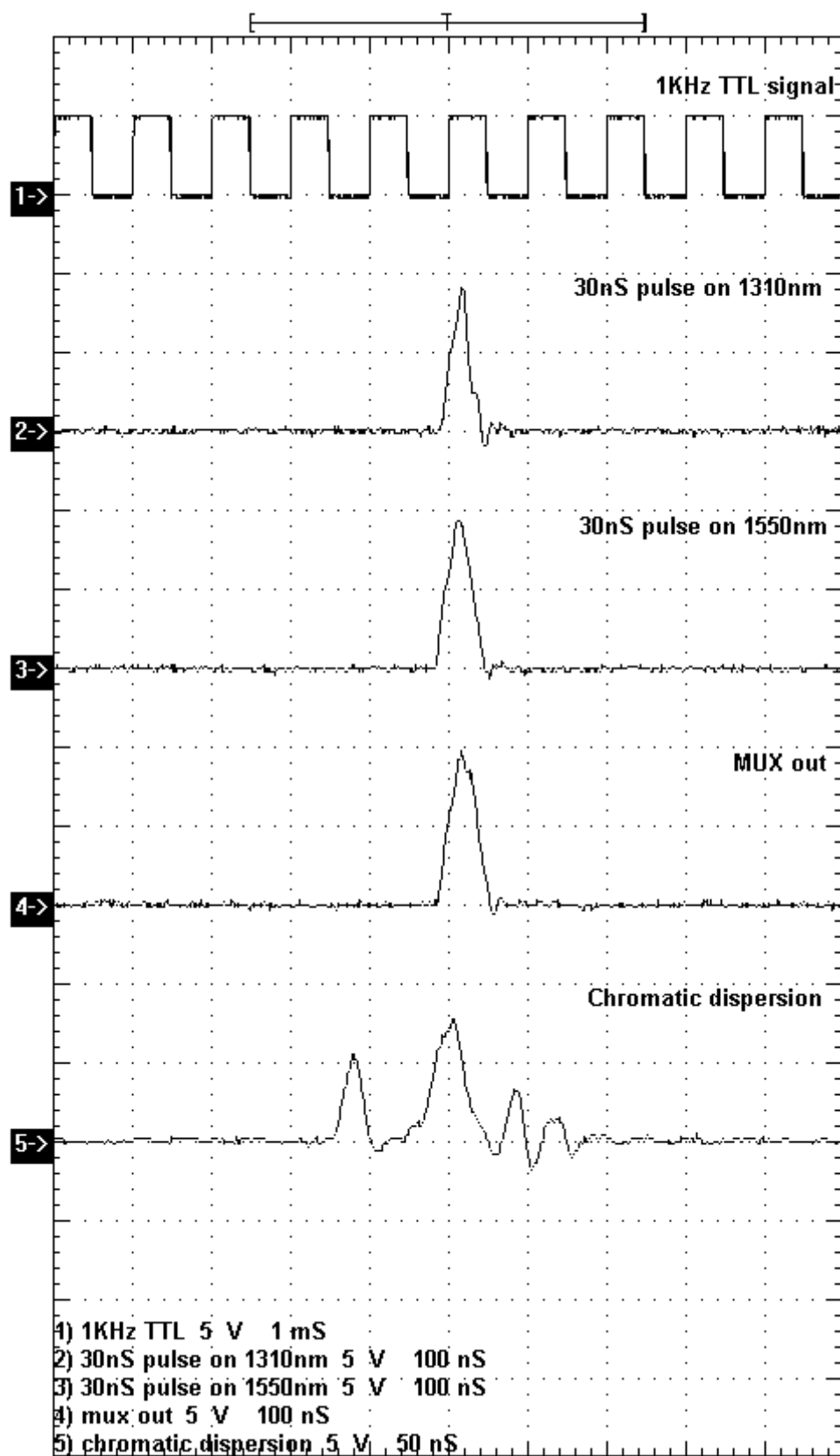
$$D = D_0 = (44 \times 10^3 \text{ ps}) / (25 \times 10^3 \times 240 \text{ nm})$$

$$= 7.333 \text{ ps/km/nm at } \lambda_0 = 1450 \text{ nm}$$

The approximate value of Dispersion at 1450nm is 7.333 ps/km/nm.

Since we have found the dispersion D_0 at the center wavelength of 1450nm and knowing that the dispersion is null at 1310nm, we calculate the dispersion at 1550nm as $7.333 \times 2 = 14.66 \text{ ps/km/nm}$

In this experiment we have assumed that D varies linearly with λ between 1310nm and 1550nm (Refer fig.1)



EXPERIMENT NO. 6

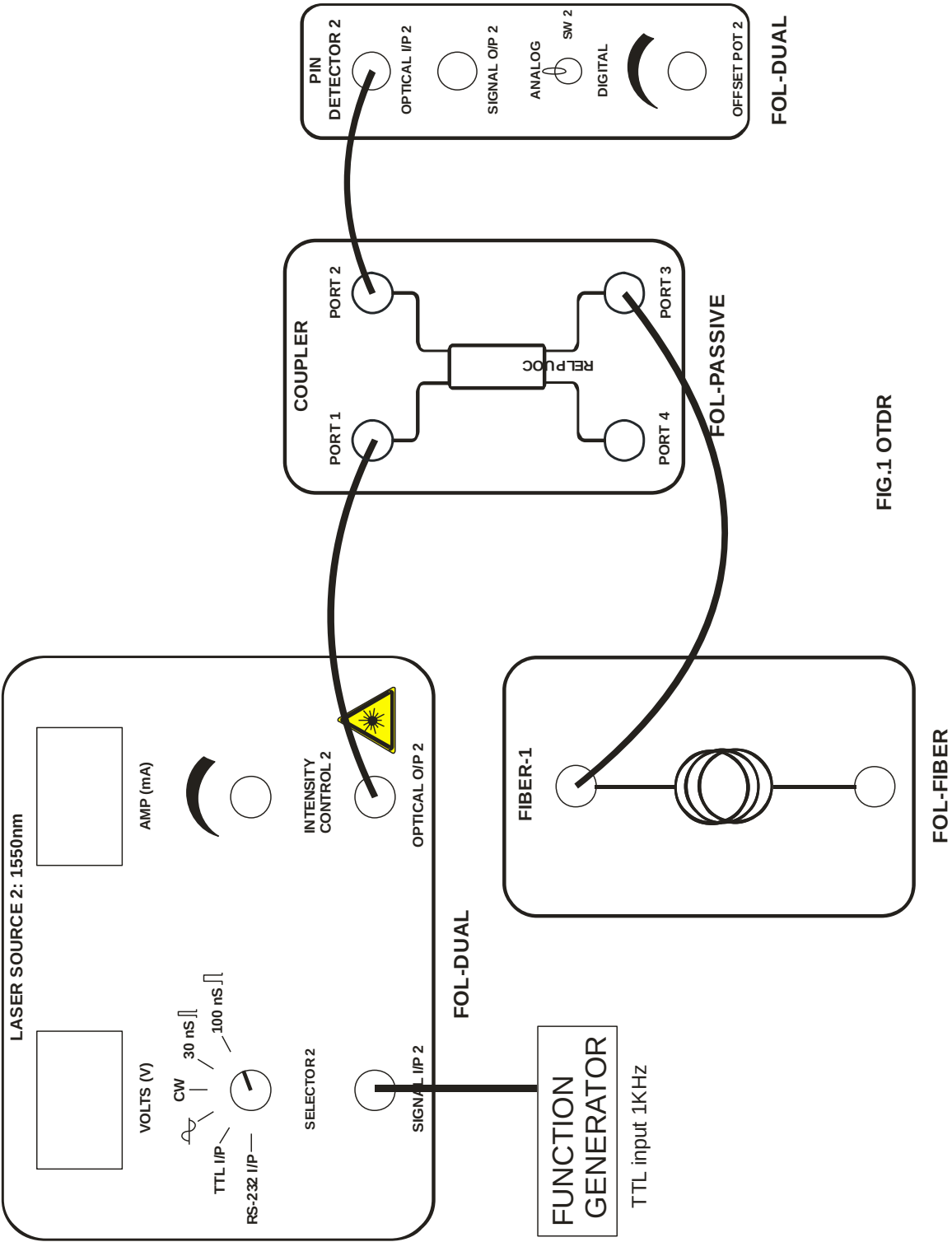


FIG.1 OTDR

EXPERIMENT NO. 6

OBJECTIVE

To study OTDR using source of 1550nm laser diode & PIN TIA Photo detector.

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE (need to be purchasedseparately)
- FOL-FIBER (need to be purchasedseparately)
- 1 meter ST-ST glass Fiber chords - 05
- Dual Channel Digital Storage Oscilloscope (DSO) 100 MHz
- Function Generator

THEORY

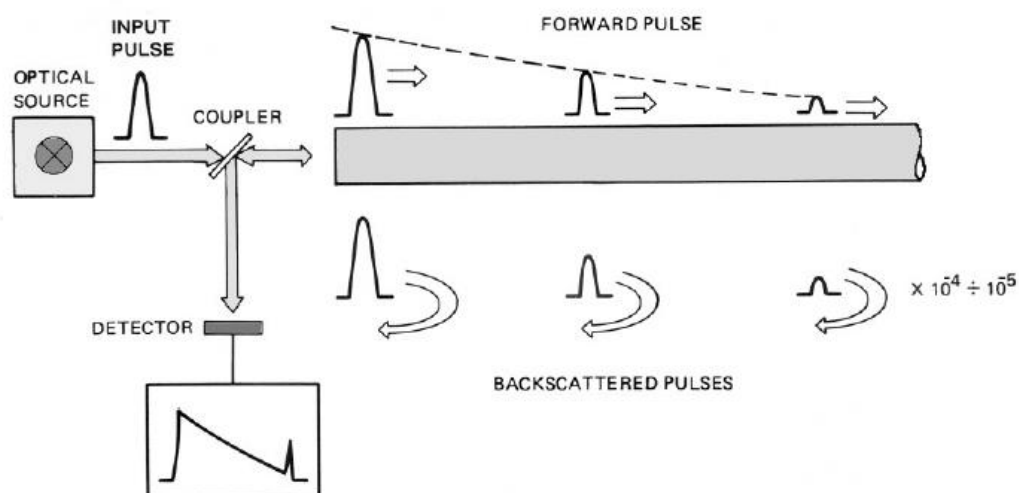
BASIC WORKING PRINCIPLE:

More accurate measurements can be made by means of an Optical Time Domain Reflectometer (OTDR). The main characteristic which makes the instrument so versatile and useful is its ability to measure the fibers local properties. The OTDR can measure not only the fiber attenuation;it can also give the trend of distributed losses along the fiber link.

For such reasons, OTDR is used in

- Loss fiber coefficient
- Splices and connectors losses
- Splices and connectors spatial position
- To find out a fiber link length (or two subsections)
- To recognize fiber defect or damages

The OTDR is based on the Rayleigh scattering, taking place in the optical fiber.



1. An optical laser source produces a pulse that is launched into the optical fiber through a directional coupler.

2. During the propagation, a portion of the pulse power is back-reflected due to the Rayleigh scattering and is collected by a detector through the directional coupler.
3. At the time τ the detector receives reflected pulse power coming from a distance:

$$R = c \Delta t / 2n$$

Where n is the fiber refractive index.

Δt = time difference between Reference pulse and back reflected pulse.

RELATIVE THEORY:

OTDR is the detection and analysis, in relation to time, of the retro diffused light by an optical fiber. The retro diffused power, in relation with time can be expressed as,

$$P_r(t) = 1/2 P_o \tau V_g \alpha_d S^{-2\alpha V_g t}$$

Where,

P_o = the average power injected into the fiber

τ = pulse length

$V_g \cong c/n_1$ = Group velocity

α_d = Rayleigh diffusion coefficient

S = retro diffusion factor

α = attenuation of the fiber

The curve representing $10 \log P_r/P_o$ in relation to $z = V_g t$ (z expressed in Km) is a straight line, whose slope is equal to $2 A_{dB}$, where A_{dB} is the linear attenuation of the fiber expressed in dB/Km.

The spatial resolution of the OTDR is related to the width of the pulse by the relation,

$$R = c \Delta t / 2n$$

Where $c = 3 \times 10^8$ m/s is the light in the vacuum.

n = effective index of a material of a fiber core (for glass, $n \approx 1.5$).

R = distance from reflected pulse power coming.

Δt = time difference between Reference pulse and back reflected pulse

ISOLATOR

Isolators are optical devices that allow light to be transmitted in one direction only. They are most often used to prevent any light reflected back down the fiber from entering the source, thus preventing any feedback problems.

The simplest type of isolator consists of a polarizer, followed by a quarter wave plate. Such a device will block simple reflections, provided that the output polarization from the quarter wave plate is not modified by other optical elements. Unfortunately, this is rarely the case in fiber optic systems. A much better type of isolator uses a polarizer with a Faraday rotator to block the return light. This type of isolator blocks all types of polarization and thus makes a much higher quality isolator. This is the type of isolator used by OZ Optics. With this type of isolator, isolation levels of 35dB can be achieved for 514 to 1064nm wavelengths and 42dB for 1300 and 1550nm wavelengths. Furthermore,

isolation levels of 60dB can be achieved for 1300nm and 1550nm by cascading two isolators together.

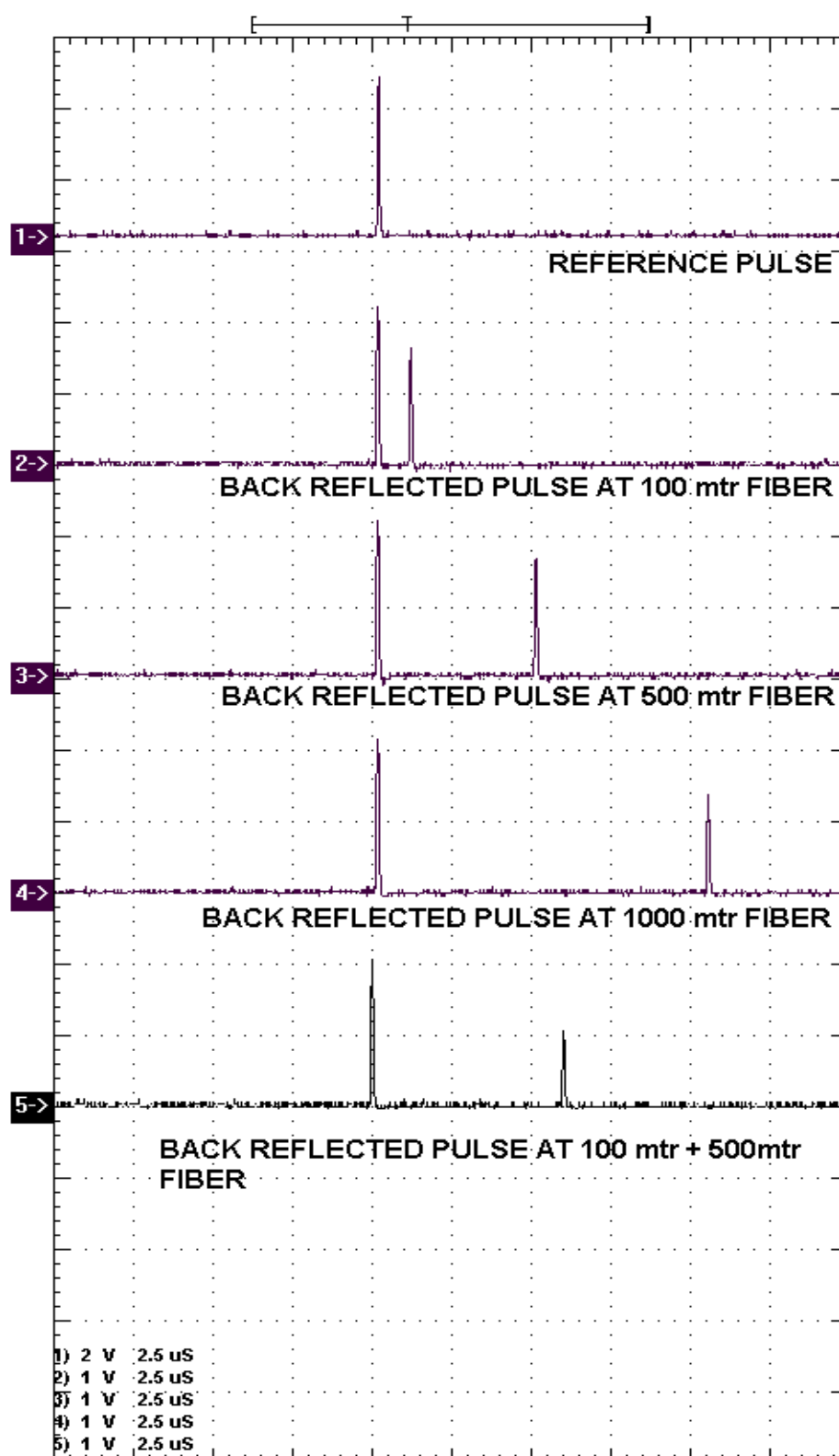
PROCEDURE

1. Refer to figure 1. Connect function generator to signal i/p 2 post of the FOL-DUAL, using the BNC connector. Set function generator to TTL i/p 1 KHz.
2. Connect Optical o/p 2 post of the FOL-DUAL module to Port-1 of Coupler (FOL-PASSIVE module), using ST-ST 1 meter SM patch cord provided with modules.
3. Connect Optical i/p 2 post of the FOL-DUAL module in the PIN TIA detector 2 section, to port-2 of the Coupler (FOM-02 module), using ST-ST 1 meter SM patch cord provided with modules.
4. Connect port -3 of the Coupler (FOL-PASSIVE module) to one end of fiber-1 (FOL-FIBER module), using ST-ST 1 meter SM patch cord provided with modules and keep the other end of the fiber as it is.
5. Connect DSO to Signal o/p 2 post (FOL-DUAL module) as an output from the detector.
6. Connect the power supply to FOL-DUAL module. While connecting this, ensure that the power supply is OFF.
7. Keep INTENSITY CONTROL2 pot on FOL-DUAL, fully anticlockwise.
8. Set external function generator to 1 KHz TTL input and connect to SIGNALI/P2 of FOL-DUAL.
9. Set SELECTOR 2 switch on 100nSec pulse mode and SW2 switch of PIN DETECTOR 2 on analog mode.
10. Now switch ON the power supply of module FOL-DUAL, function generator and Digital storage Oscilloscope.
11. Use OFFSET POT 2 to adjust the output.
12. Observe output at SIGNAL O/P 2 post on front panel of FOL-DUAL using DSO. You will observe one transmitter pulse and back reflected pulse, measure the time delay between these two pulses this will be Δt
13. Repeat the procedure using fiber-2 and observe the waveform using DSO. Note down time Δt between two pulses. Find the length of the fiber.
14. Repeat the procedure using fiber-3 and observe the waveform using DSO. Note down time Δt between two pulses. Find the length of fiber.
15. Repeat the procedure using combination of fiber-1 and fiber-2 and observe the waveform using DSO. Note down time Δt between two pulses. Find the length of fiber.
16. Repeat the procedure using combination of fiber-1, fiber-2 and fiber-3 and observe the waveform using DSO. Note down time Δt between two pulses. Find the length of fiber.

Note : We have provided three different types of fiber , you can vary the length in following manner.

100m, 500m, 1000m, 600m, 1100m, 1500m, 1600m .

Captured wave forms for fiber



CALCULATION

Use
the formula,

$$R = c \Delta t / 2n$$

Where $c = 3 \times 10^8$ m/s is the light in the vacuum.

n = effective index of a material of a fiber core (for glass, $n \approx 1.5$).

R = distance from reflected pulse power coming.

Δt = time difference between Reference pulse and back reflected pulse observed in DSO.

Distance from reflected pulse power (R) calculated for 100m, 500m & 1000m is as follows,

1) When 100m Fiber used

$$\Delta t = 1.020 \mu s$$

$$n = 1.5 \quad \text{then}$$

$$R = c \Delta t / 2n$$

$$= (3 \times 10^8 \times 1.020 \mu) / (2 \times 1.5)$$

$$= 100 \text{ m}$$

$R = 100 \text{m}$ indicate that the reflection of pulse takes place from 100m distance

2) When 500m Fiber used

$$\Delta t = 4.960 \mu s$$

$$n = 1.5 \quad \text{then}$$

$$R = c \Delta t / 2n$$

$$= (3 \times 10^8 \times 4.960 \mu) / (2 \times 1.5)$$

$$= 496 \text{ m}$$

$R = 496 \text{ m}$ indicate that the reflection of pulse takes place from 496m distance

3) When 1000m Fiber used

$$\Delta t = 9.680 \mu$$

$$n = 1.5 \quad \text{then}$$

$$R = c \Delta t / 2n$$

$$= (3 \times 10^8 \times 9.680 \mu) / (2 \times 1.5)$$

$$= 968 \text{ m}$$

$R = 968 \text{m}$ indicate that the reflection of pulse takes place from 968m distance

Similarly, we can do the calculation for distance from reflected pulse at different fiber combination.

CONCLUSION

By using OTDR principle of Rayleigh scattering we can find out the distance, to recognize fiber breakages, defects and damages caused.

EXPERIMENT NO. 7

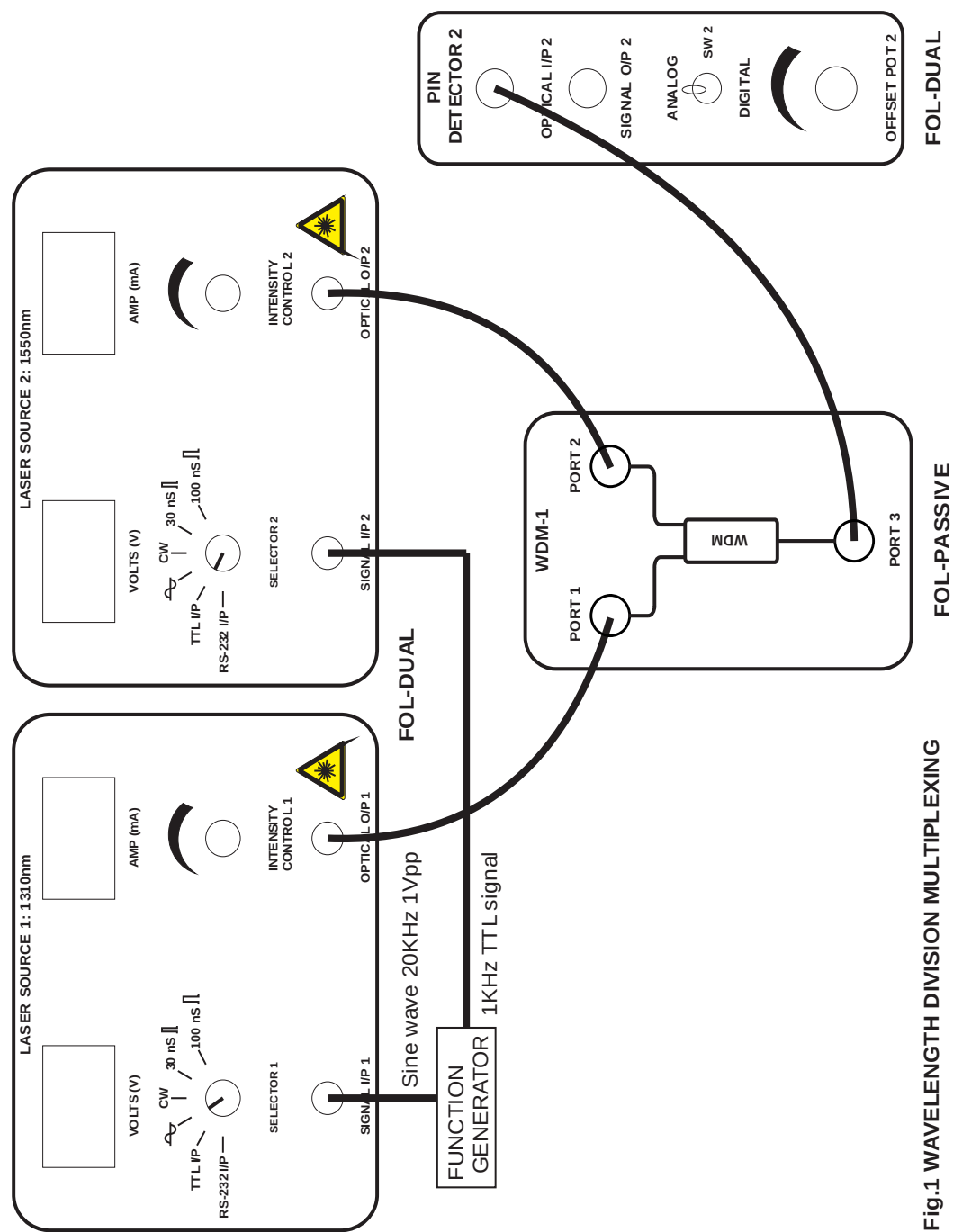


Fig.1 WAVELENGTH DIVISION MULTIPLEXING

EXPERIMENT NO. 7

OBJECTIVE

The objective of this experiment is to study Fiber Optics wavelength division.

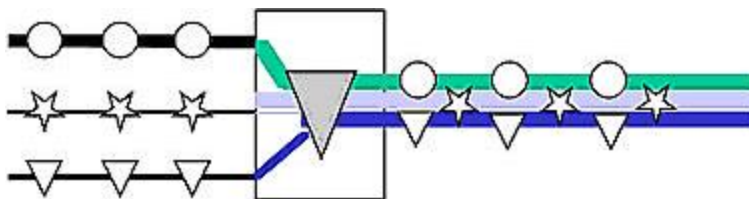
EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- 1 meter ST-ST glass Fiber patch chords - 05
- Dual Channel Digital Storage Oscilloscope (DSO) 100 MHz
- Function Generator 2 MHz– (02 nos or dual channel)

THEORY

In fiber-optic communications, wavelength-division multiplexing (WDM) is a technology which multiplexes multiple optical carrier signals on a single optical fiber by using different wavelengths (colours) of laser light to carry different signals. This allows for a multiplication in capacity, in addition to making it possible to perform bidirectional communications over a single strand of fiber.

The term wavelength-division multiplexing is commonly applied to an optical carrier (which is typically described by its wavelength).



- Merges optical traffic onto one optical fiber
- Allows high flexibility in expanding bandwidth
- Not dependant on signal, bit rate or format

Wavelength Division Multiplexing (WDM) applies to any fiber optic system where two or more signals are being transmitted, at different optical wavelengths, over a single fiber. This type of transmission increases the capacity of the optical fiber. Under WDM, the optical transmission spectrum is carved up into a number of non-overlapping wavelengths (or frequencies) with each supporting a single communication channel.

WDM technology utilizes a multiplexer that accepts data from the end user's equipment and transmits this data, via lasers, onto a channel. The multiplexer simultaneously supports many end users in this way. The multiplexer then combines all of the channels into one signal and sends that signal, over one optic fiber, to another device at the destination site called a de-multiplexer. The de-multiplexer separates the signal back into channels of different frequencies and passes those channels on to their assigned destinations.

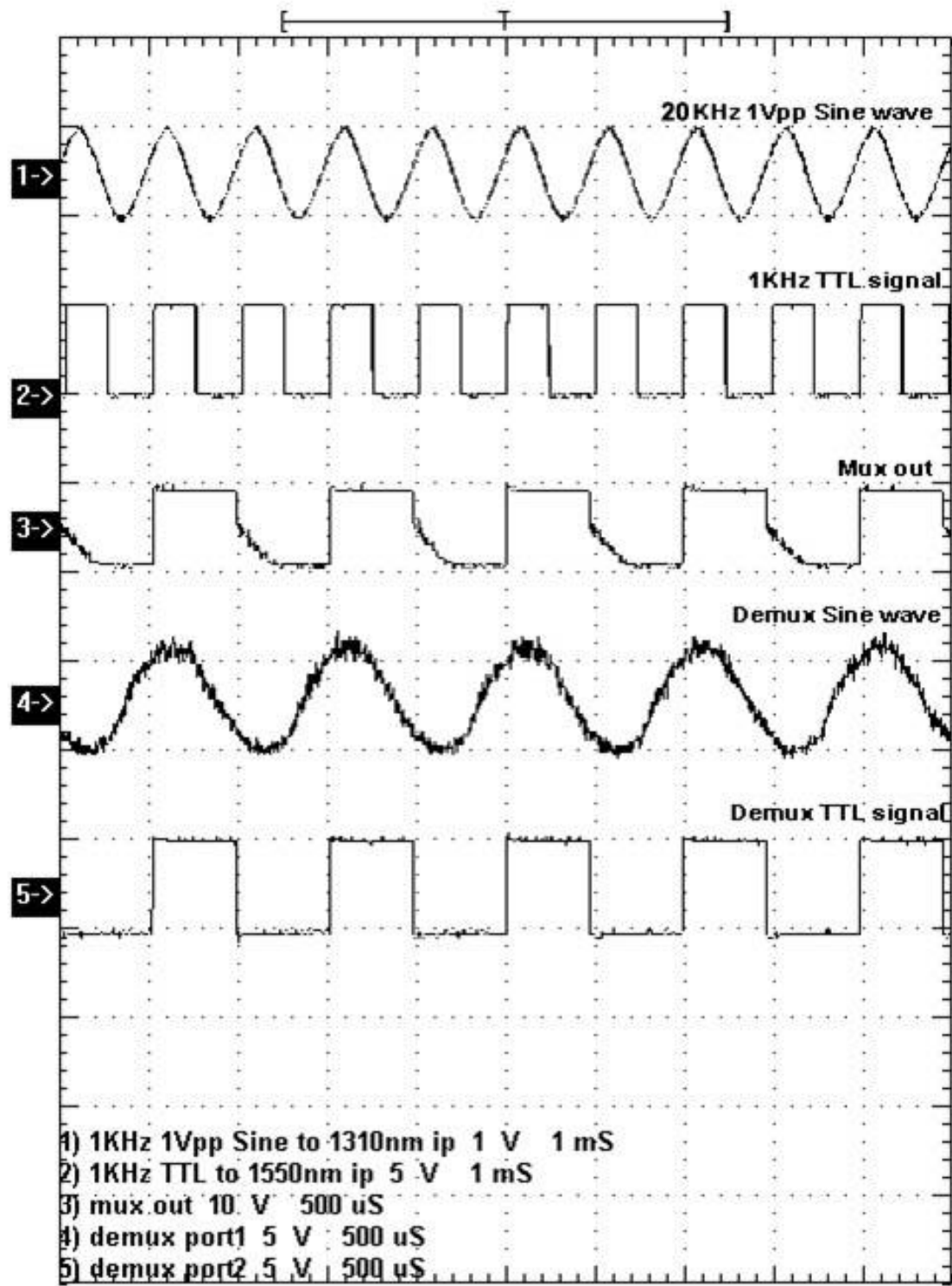
PROCEDURE

WAVELENGTH DIVISION MULTIPLEXING / DEMULTIPLEXING

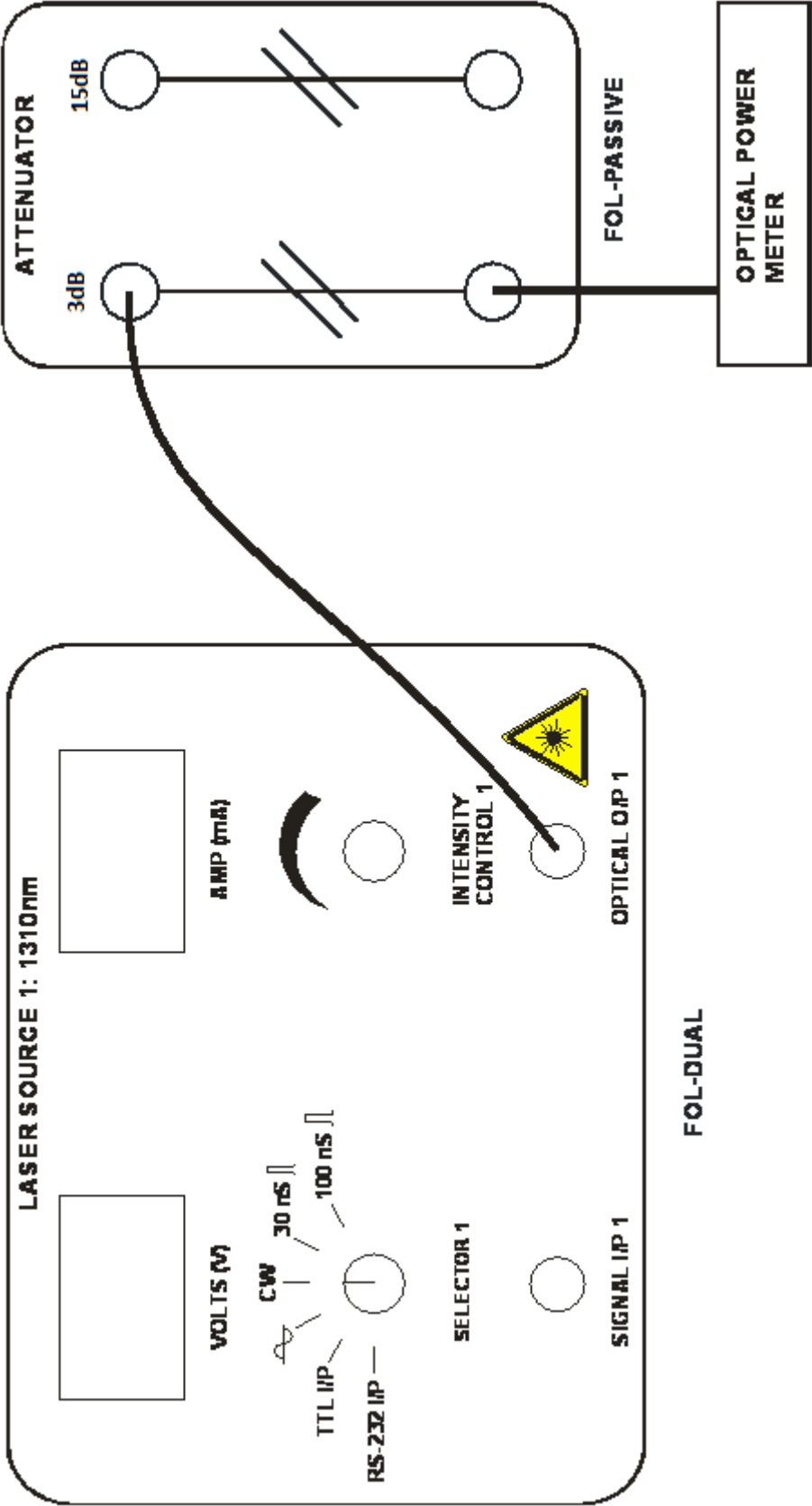
1. Refer to figure 1. Set up the Function Generator for o/p with TTL 1 KHz Signal to the SIGNAL I/P2post and set other function generator for 20KHz, 1Vp-p sine wave Signal to the SIGNAL I/P 1 post of FOL-DUAL, using BNC-BNC cable.
2. Individual check for establish Analog link for 1550nm with PIN DETECTOR 2 and Digital link 1310nm with DETECTOR 1. This is just a verification process , once this is done connect the optical signal of 1310nm to PORT-1 of WDM-1 and 1550nm to PORT-2 of WDM-1.
3. SWITCH OFF the FOL-DUAL system.
4. Now connect Optical o/p 1 post (FOL-DUAL module) to Port1 of WDM 1 on FOL-PASSIVE module, using ST-ST 1 meter SM patch cord provided with modules.
5. Connect Optical o/p 2 post (FOL-DUAL module) to Port2 of WDM 1 on FOL-PASSIVE module, using ST-ST 1 meter SM patch cord provided with modules.
6. Connect Port 3 of WDM on FOL-PASSIVE to optical input2 post on FOL-DUAL module, using ST-ST 1 meter SM patch cord provided with modules.
7. Connect CH1 of DSO to Signal output 2 post (FOL-DUAL module).
8. Keep SW2 switch on Analog mode.
9. Connect the mains power cable to FOL-DUAL module at the back. While connecting this, ensure that the power supply is OFF.
10. Now switch ON the power supply of module FOL-DUAL, function generator and the Digital storage Oscilloscope.
11. Check and observe multiplexed output on signal output 2 post of PIN DETECTOR 2. Use OFFSET POT 2 to adjust the output.
12. Now connect WDM 1 Port3 output to Port3 of WDM 2 in FOL-PASSIVE.
13. Connect Port1 of WDM 2 to Optical i/p 2 of Detector 2 in FOL-DUAL.
14. Observe the de-multiplexed Sine wave on Signal o/p 2 in FOL-DUAL.
15. Adjust the Offset pot 2 if required.
16. Switch SW2 on Analog mode while checking Analog output.
17. Connect Port 2 of WDM 2 in FOL-PASSIVE to Optical i/p 2 of Detector 2 in FOL-DUAL.
18. Put switch SW2 to Digital position in FOL-DUAL.
19. Observe the de-multiplexed TTL signal on Signal o/p 2 post of FOL-DUAL.
20. If required adjust the Offset pot 2 in FOL-DUAL.

CONCLUSION

Thus, using Fiber Optic WDM2x1, 1310nm and 1550nm are multiplexed and using another WDM1x2, de-multiplexing is performed. The respective waveforms are observed on DSO.



EXPERIMENT NO. 8



BLOCK DIAGRAM FOR MEASUREMENT OF ATTENUATION IN FIBER OPTIC ATTENUATOR

EXPERIMENT NO. 8

OBJECTIVE

To Measure the Attenuation in Fiber Optic Attenuator

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- 1 meter ST-ST glass Fiber patch chords
- Optical Power Meter

THEORY

In certain applications we may need to use calibrated attenuators in order to reduce the luminescent intensity of a known quantity. The fiber attenuators are easy to realize. They are made by realizing “bad splicing” between two fibers. The two cores of the two identical fibers are voluntarily spliced together transversely.

Optical attenuators used in fiber optic communications systems may use a variety of principles for their functioning. Those using the gap-loss principle are sensitive to the modal distribution ahead of the attenuator, and should be used at or near the transmitting end, or they may introduce less loss than intended. Optical attenuators using absorptive or reflective techniques avoid this problem.

The basic types of optical attenuators are fixed, step-wise variable, and continuously variable.

PROCEDURE

1. Make sure that the Intensity control 1 and Intensity Control 2 pots are turned fully anticlockwise before switching on the FOL-DUAL module.
2. Keep the Selector 1 pot in FOL-DUAL module on CW selection.
3. Connect the optical power meter to the Optical O/P 1 post of the Laser Source: 1310nm on the FOL-DUAL module.
4. Gradually turn the Intensity Control 1 Pot clockwise till you get around 1.1mW of optical power reading on the Optical Power meter. Note down this as power P_1 .
5. Now remove the power meter and connect Optical O/P 1 post to one of the posts of the 3dB Attenuator on the FOL-PASSIVE module.
6. Connect the optical power meter to remaining post of the 3dB attenuator.
7. Note down the reading as power P_2 and find out the Attenuation loss,
8. $A_{dB} = 10 \log (P_2 / P_1)$ where, P_2 is the Optical power after attenuator and P_1 is the Laser source power input to Attenuator.
9. Repeat the measurement after altering the posts of the Attenuator. Thus deduce the Attenuation A_{dB} in both the directions.
10. Repeat the experiment for 15dB attenuator and 1550nm Laser source and calculate the attenuation Loss A_{dB} .

MEASUREMENTS

Example:

P_1 (1310nm Laser source power input to 3dB Attenuator) = 1.1mW

P_2 (Optical power after 3dB attenuator) = 491.35uW

Attenuation loss for 3dB attenuator,

$$A_{dB} = 10 \log (P_2 / P_1)$$

$$= -3.5dB$$

Example:

P_1 (1310nm Laser source power input to 15dB Attenuator) = 1.1mW

P_2 (Optical power after 15dB attenuator) = 31.00uW

Attenuation loss for 15dB attenuator,

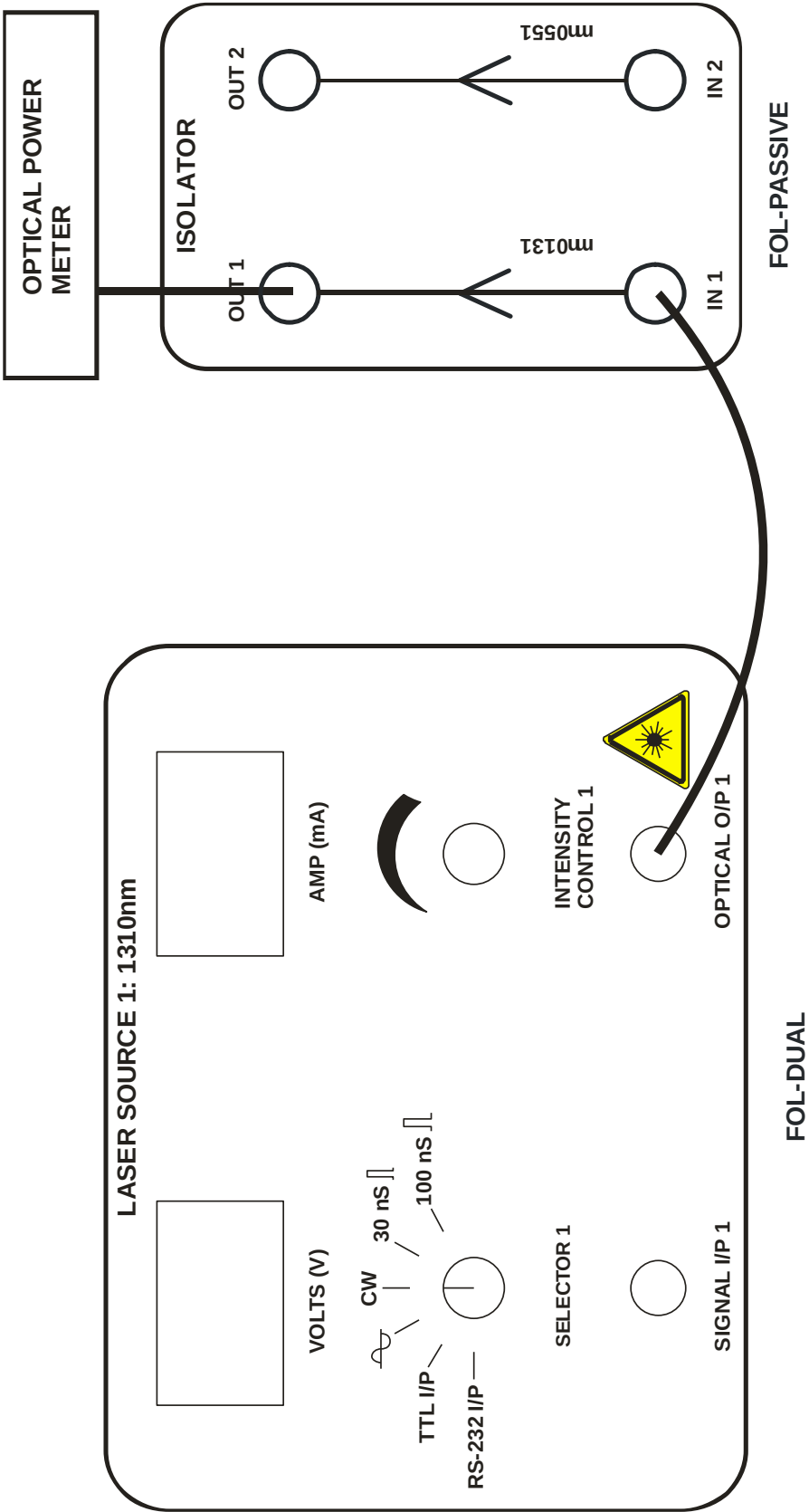
$$A_{dB} = 10 \log (P_2 / P_1)$$

$$= -15.5dB$$

CONCLUSION

The Attenuation for 5dB and 10dB attenuators is verified with 1310nm and 1550nm Laser source.

EXPERIMENT NO. 9



BLOCK DIAGRAM FOR MEASUREMENT OF INSERTION LOSSES IN FIBER OPTIC ISOLATOR

EXPERIMENT NO. 9

OBJECTIVE

To Measure the Insertion Losses & Isolation Rate in Fiber Optic Isolator.

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- 1 meter ST-ST glass Fiber patch chords
- Optical Power Meter

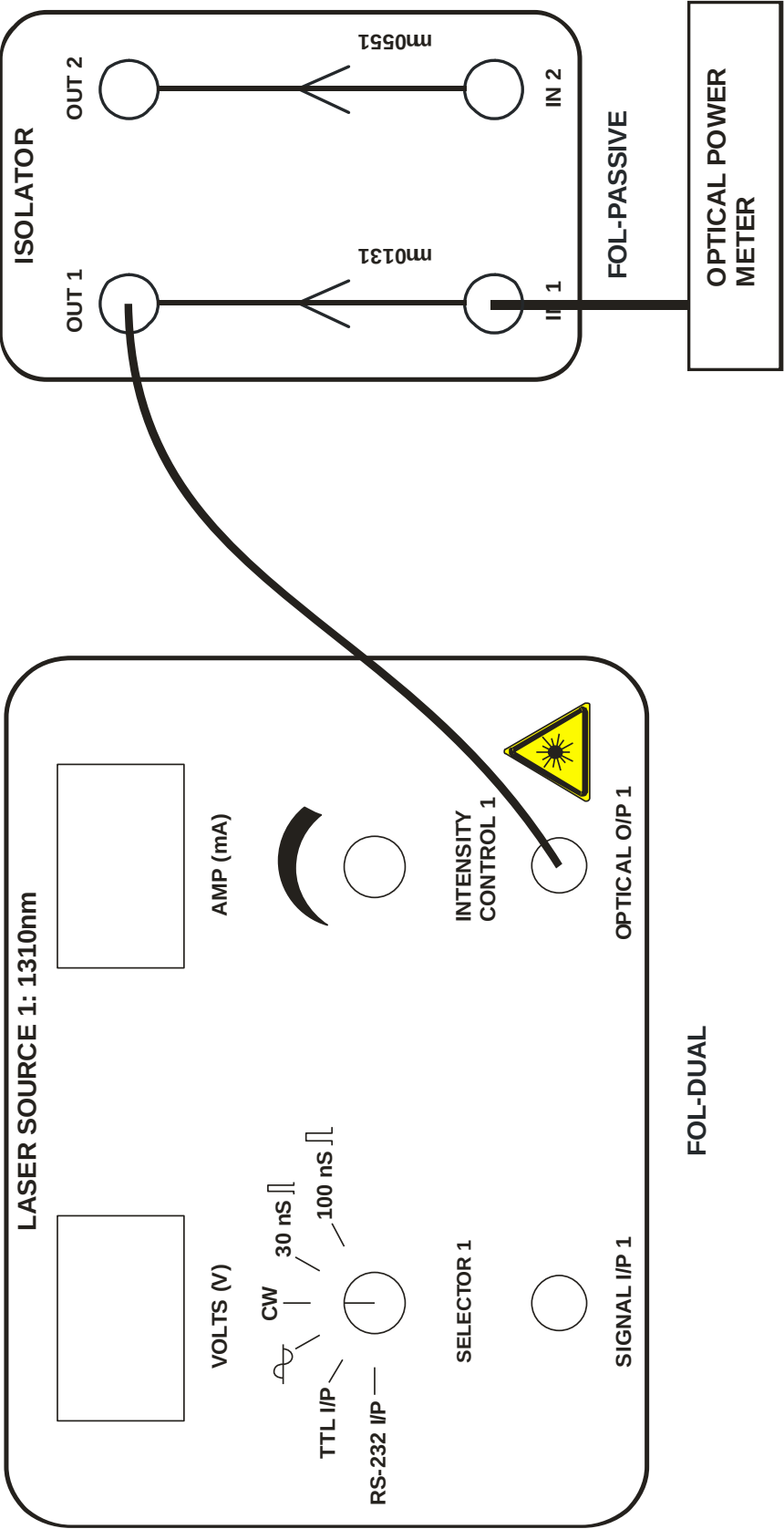
THEORY

An optical isolator, or optical diode, is an optical component which allows the transmission of light in only one direction. They are typically used to prevent unwanted feedback into an optical oscillator, such as a laser cavity. The operation of the device depends on the Faraday effect (which in turn is produced by magneto-optic effect), which is used in the main component, the Faraday rotator.

The main component of the optical isolator is the Faraday rotator. The magnetic field, $\mathbf{B}$, applied to the Faraday rotator causes a rotation in the polarization of the light due to the Faraday Effect. The angle of rotation, β , is given by,

$$\beta = \nu B d$$

Where, ν is the Verdict constant of the material (amorphous or crystalline; solid, liquid, or gaseous) of which the rotator is made, and d is the length of the rotator. Specifically for an optical isolator, the values are chosen to give a rotation of 45 degrees.



BLOCK DIAGRAM FOR MEASUREMENT OF ISOLATION RATE IN FIBER OPTIC ISOLATOR

PROCEDURE

1. Make sure that the Intensity control 1 and Intensity Control 2 pots are turned fully anticlockwise before switching on the FOL-DUAL module.
2. Keep the Selector 1 pot in FOL-DUAL module on CW selection.
3. Connect the optical power meter to the Optical O/P 1 post of the Laser Source: 1310nm on the FOL-DUAL module.
4. Gradually turn the Intensity Control 1 Pot clockwise till you get around 1.3mW of optical power reading on the Optical Power meter. Note down this as power P_1 .
5. Now remove the power meter and connect Optical O/P 1 post to input post of the isolator in FOL-PASSIVE module.
6. Connect the optical power meter to output post of the Isolator.
7. Note down the reading as power P_2 and find out the Insertion Losses,
 - a. $\alpha_{dB} = 10 \log (P_2 / P_1)$. Where, P_2 is the Optical output power after Isolator and P_1 is the Laser source power input to Isolator.
8. Connect the optical power meter to the Optical O/P 1 post of the Laser Source: 1310nm on the FOL-DUAL module.
9. Gradually turn the Intensity Control 1 Pot clockwise till you get around 1.7mW of optical power reading on the Optical Power meter. Note down this as power P_3 .
10. Now connect the Optical O/P 1 post to output post of the isolator in FOL-PASSIVE module.
11. Connect the optical power meter to Input post of the Isolator.
12. Note down the reading as power P_4 .
13. Calculate the Isolation rate, $I_{dB} = 10 \log (P_3 / P_4)$.
14. Repeat the measurements for 1550nm Laser source.

MEASUREMENTS

Example:

P_1 (1310nm Laser source power input to Isolator input) = 1.3mW

P_2 (Optical power at Isolator output) = 1.1mW

Insertion losses are,

$$\alpha_{dB} = 10 \log (P_2 / P_1) \\ = - 0.72dB$$

P_3 (1310nm Laser source power input to Isolator output) = 1.7mW

P_4 (Optical power at Isolator input) = 150 μ W

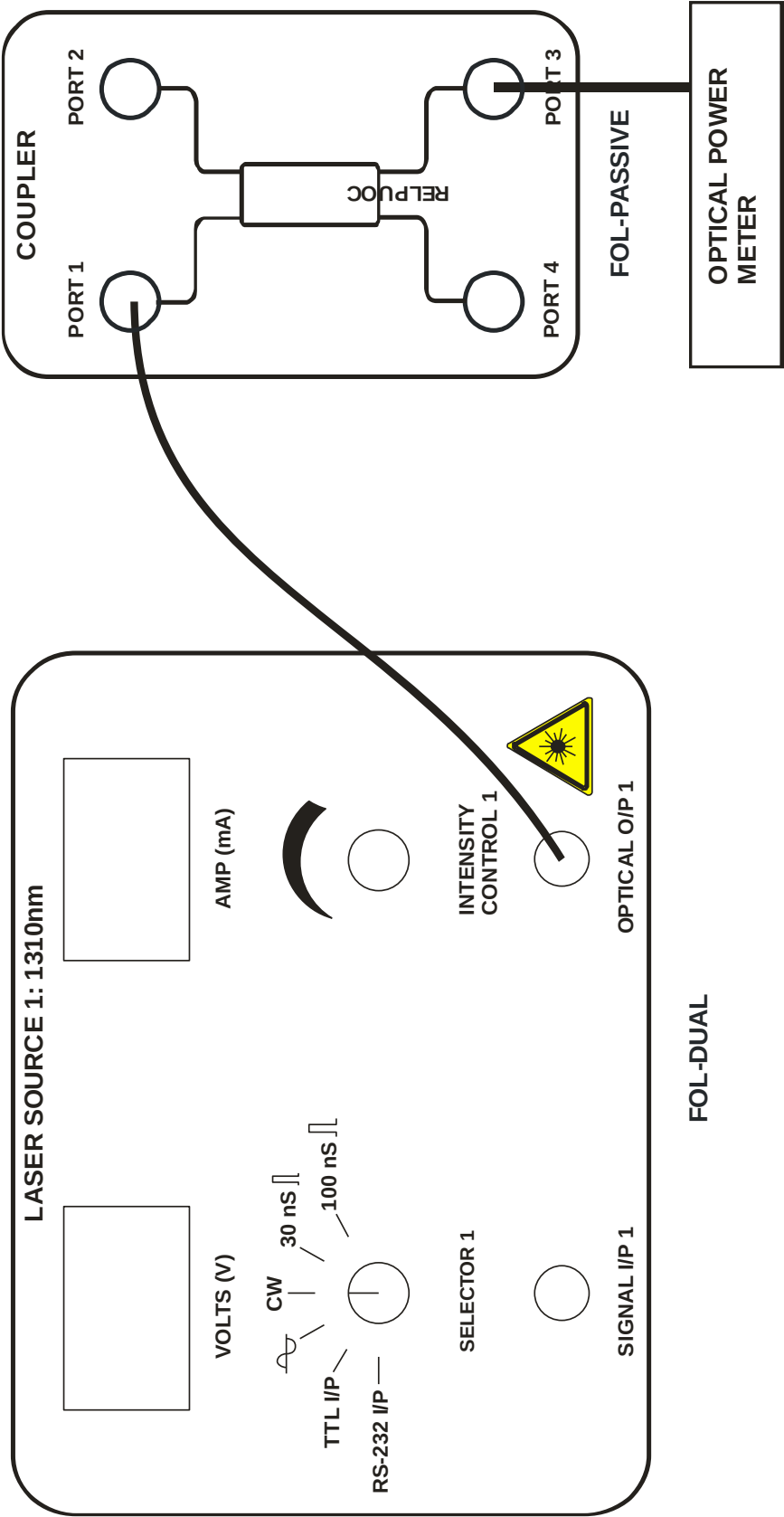
Isolation Rate is,

$$I_{dB} = 10 \log (P_4 / P_3) \\ = - 40dB$$

CONCLUSION

We note that the Isolators are very efficient on the operating wavelength as they have low insertion losses (0.72dB for example) and an isolation rate that is higher than 40dB.

EXPERIMENT NO. 10



BLOCK DIAGRAM FOR MEASUREMENT OF INSERTION LOSSES & COUPLING COEFFICIENT IN FIBER OPTIC COUPLER

EXPERIMENT NO. 10

OBJECTIVE

To Measure the Insertion Losses & Coupling Coefficient in Fiber Optic Coupler

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- 1 meter ST-ST glass Fiber patch cords
- Optical Power Meter(need to be purchase separately)

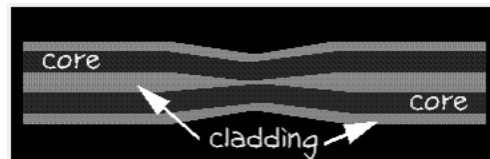
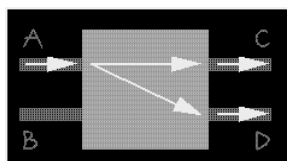
THEORY

An optical coupler is a device that splits the light.

There are several types of couplers:

- Directional coupler
- Star coupler

The four-port directional coupler, also denoted 2 x 2 coupler, is the simplest coupler.



The light arrives for instance at port A and is split between port C and D. In the most common case, 50% of the light power will go in C and D. Optical couplers are made by fusing and tapering two fibers together so that the cores are close enough to each other for optical coupling to occur.

In practice,

- twist two fibers together
- Heat the in a furnace: the fibers are softened and they fuse.

The principle of optical couplers varies whether the fibers used are single mode or multimode.

PROCEDURE

1. Make sure that the Intensity control 1 and Intensity Control 2 pots are turned fully anticlockwise before switching on the FOL-DUAL module.
2. Keep the Selector 1 pot in FOL-DUAL module on CW selection.
3. Connect the optical power meter to the Optical O/P 1 post of the Laser Source: 1310nm on the FOL-DUAL module.
4. Gradually turn the Intensity Control 1 Pot clockwise till you get around 1.7mW of optical power reading on the Optical Power meter. Note down this as power P_1 .
5. Now remove the power meter and connect Optical O/P 1 post to Port 1 of the Coupler in FOL-PASSIVE module.

6. Connect the optical power meter to Port 3 of the Coupler.
 - a. Note down the reading as power P_3 .
7. Connect the optical power meter to Port 4 of the Coupler.
8. Note down the reading as power P_4 . Find out the Insertion Losses,

$$\alpha_{dB} = 10 \log \{(P_3 + P_4) / P_1\}.$$

Where, P_3 is the Optical output power on Port 3 of the Coupler,
 P_4 is the Optical output power on Port 4 of the Coupler,
 And P_1 is the Laser source power input to Port 1 of a Coupler.
9. Calculate the Coupling Coefficient, $K = \{P_3 / (P_3 + P_4)\}$.
10. Repeat the measurements for 1550nm Laser source.

MEASUREMENTS

Example: P_1 (1310nm Laser source power input to Isolator input) = 1.7mW
 P_3 (Optical output power on Port 3 of the Coupler) = 700 μ W
 P_4 (Optical output power on Port 4 of the Coupler) = 640 μ W
 Insertion losses are,

$$\alpha_{dB} = 10 \log \{(P_3 + P_4) / P_1\}$$

$$= -1.03dB$$

Coupling Coefficient,

$$K = \{P_3 / (P_3 + P_4)\}$$

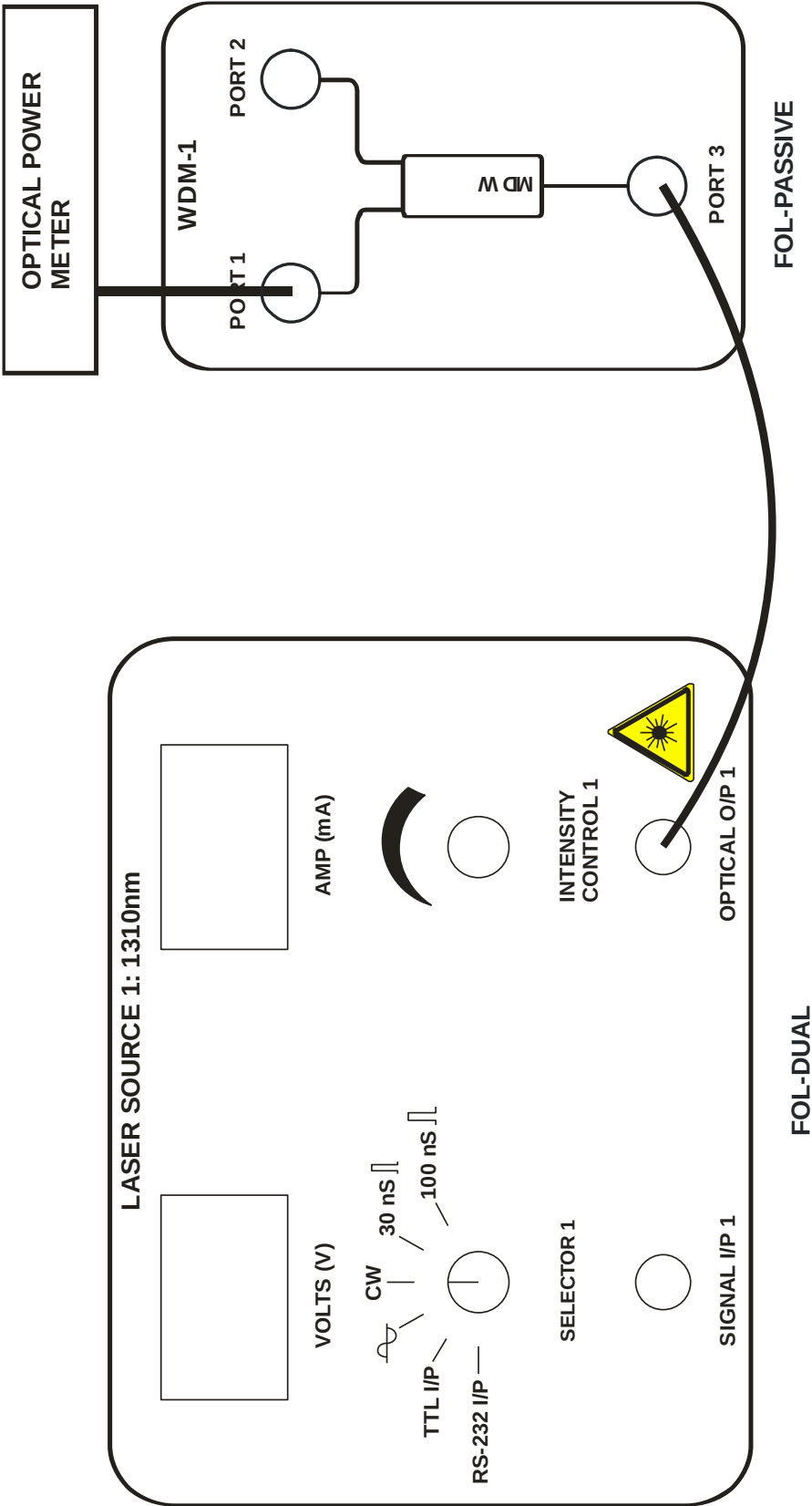
$$= 0.52$$

$K\% = 52\%$

CONCLUSION

- The couplers are symmetric, that is, that the coupling is assured no matter which path is used for the injection.
- The couplers are “wide band”, that is to say that they have more or less the same characteristics on the two wavelengths.
- Coupler used is proved to be 50/50 coupler.

EXPERIMENT NO. 11



BLOCK DIAGRAM FOR MEASUREMENT OF INSERTION LOSSES & COUPLING COEFFICIENT IN FIBER OPTIC MULTIPLEXER

EXPERIMENT NO. 11

OBJECTIVE

To Measure the Insertion Losses & Coupling Coefficient in Fiber Optic Multiplexer

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- 1 meter ST-ST glass Fiber chords
- Optical Power Meter

THEORY

In fiber-optic communications, **wavelength-division multiplexing (WDM)** is a technology which multiplexes multiple optical carrier signals on a single optical fiber by using different wavelengths of laser light to carry different signals. This allows for a multiplication in capacity, in addition to enabling bidirectional communications over one strand of fiber. This is a form of frequency division multiplexing (FDM) but is commonly called wavelength division multiplexing.

The term wavelength-division multiplexing is commonly applied to an optical carrier (which is typically described by its wavelength), whereas frequency-division multiplexing typically applies to a radio carrier (which is more often described by frequency).

As the coupling coefficient K of a coupler depends on the wavelength, we can make a coupler whose coupling coefficient is $K=0$ for a wavelength λ_1 and $K=1$ for another wavelength λ_2 .

PROCEDURE

1. Make sure that the Intensity control 1 and Intensity Control 2 pots are turned fully anticlockwise before switching on the FOL-DUAL module.
2. Keep the Selector 1 pot in FOL-DUAL module on CW selection.
3. Connect the optical power meter to the Optical O/P 1 post of the Laser Source: 1310nm on the FOL-DUAL module.
4. Gradually turn the Intensity Control 1 Pot clockwise till you get around 1.7mW of optical power reading on the Optical Power meter. Note down this as power P_1 .
5. Now remove the power meter and connect Optical O/P 1 post to Port 3 of the WDM 1 in FOL-PASSIVE module.
6. Connect the optical power meter to Port 1 of the WDM 1. Note down the reading as power P_4 .
7. Connect the optical power meter to Port 2 of the WDM 1. Note down the reading as power P_3 .
8. Find out the Insertion Losses,

$$\alpha_{dB} = 10 \log \{(P_3 + P_4) / P_1\}.$$

Where, P_3 is the Optical output power on Port 2 of the WDM 1,
 P_4 is the Optical output power on Port 1 of the WDM 1,
And P_1 is the Laser source power input to Port 3 of a WDM 1.

9. Calculate the Coupling Coefficient, $K = \{P_3 / (P_3 + P_4)\}$.
10. Repeat the experiment and measurements for 1550nm Laser source.
11. Repeat the experiments and measurements with WDM 2 in FOL-PASSIVE module.

MEASUREMENTS

Example 1:

P_1 (1310nm Laser source power input to Multiplexer port 3) = 1.7mW

P_3 (Optical output power on Port 2 of the Coupler) = $4.7\mu\text{W}$

P_4 (Optical output power on Port 1 of the Coupler) = 1.48mW

Insertion losses are,

$$\alpha_{\text{dB}} = 10 \log \{(P_3 + P_4) / P_1\}$$
$$= -0.59\text{dB}$$

Coupling Coefficient,

$$K = \{P_3 / (P_3 + P_4)\}$$
$$= 0.0034$$

$$K\% = 0.34\%$$

Example 2:

P_1 (1550nm Laser source power input to Multiplexer port 3) = 1.0mW

P_3 (Optical output power on Port 2 of the Coupler) = $890\mu\text{W}$

P_4 (Optical output power on Port 1 of the Coupler) = $14\mu\text{W}$

Insertion losses are,

$$\alpha_{\text{dB}} = 10 \log \{(P_3 + P_4) / P_1\}$$
$$= -0.44\text{dB}$$

Coupling Coefficient,

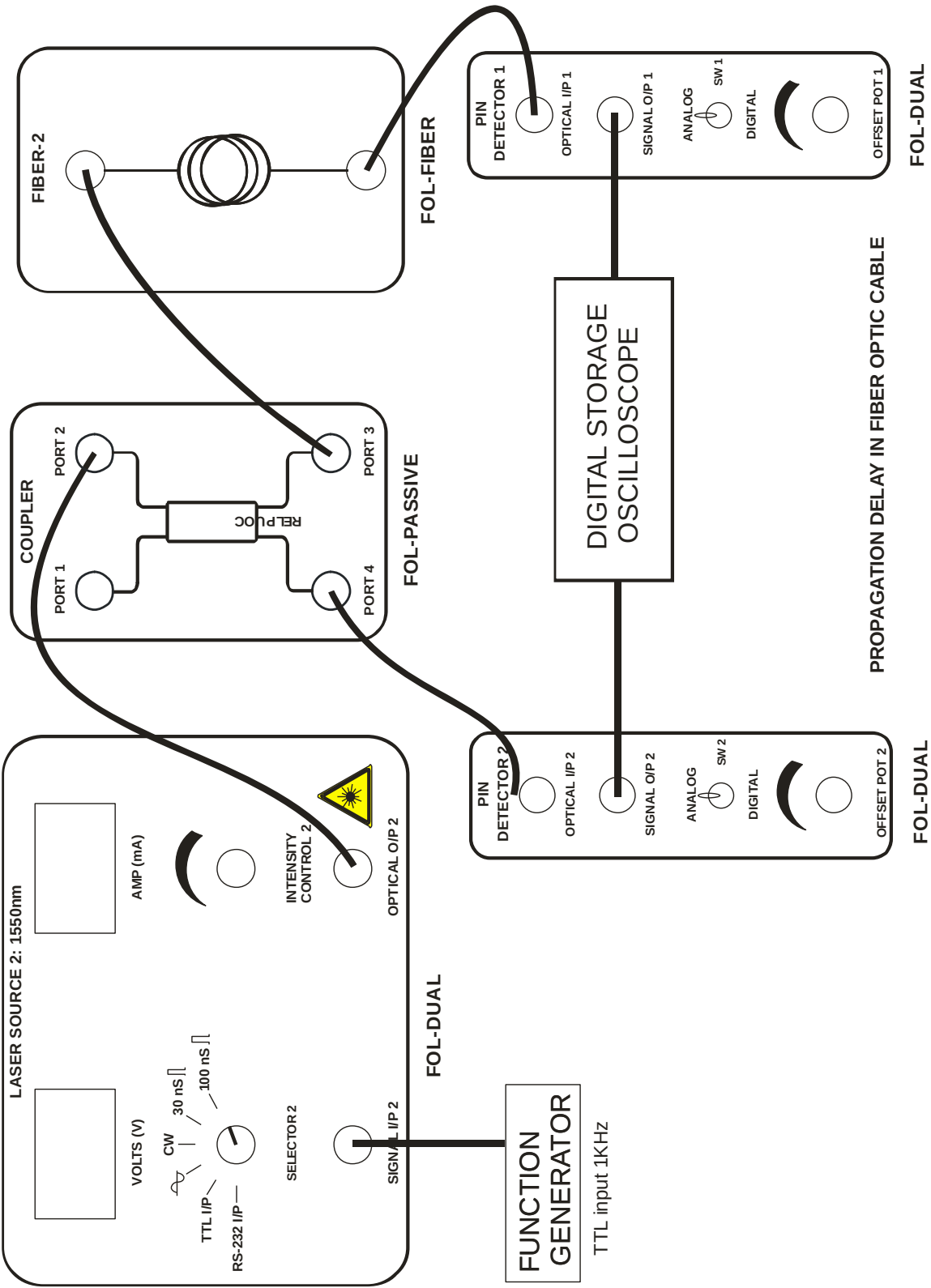
$$K = \{P_3 / (P_3 + P_4)\}$$
$$= 0.9845$$

$$K\% = 98.45\%$$

CONCLUSION

- The multiplexing function can be assured between the wavelengths 1310nm and 1550nm as this multiplexer has a coupling coefficient of about 0% at 1310nm and about 100% at 1550nm.

EXPERIMENT NO. 12



EXPERIMENT NO. 12

OBJECTIVE

To Measure the Propagation Time delay in Single Mode Fiber Optic Cable

EQUIPMENTS

- FOL-DUAL
- FOL-PASSIVE
- FOL-FIBER
- 1 meter ST-ST glass Fiber patch chords
- Optical Power Meter

THEORY

Propagation Delay—this is the time it takes for a signal to travel from one point to another over a transmission channel.

The propagation delay is equivalent to the amount of time that passes between when a signal is transmitted and when it is received on the other end of a cabling channel. The effect is akin to the delay in time between when lightning strikes and thunder is heard—except that electrical signals travel much faster than sound. The actual delay value for twisted-pair cabling is a function of the nominal velocity of propagation (NVP), length and frequency.

NVP varies according to the dielectric materials used in the cable and is expressed as a percentage of the speed of light. For example, most category 5 polyethylene (FRPE) constructions have NVP ranges from 0.65c to 0.70c (where "c" represents the speed of light $\sim 3 \times 10^8$ m/s) when measured on finished cable. Teflon (FEP) cable constructions range from 0.69c to 0.73c, whereas cables made of PVC are in the 0.60c to 0.64c range.

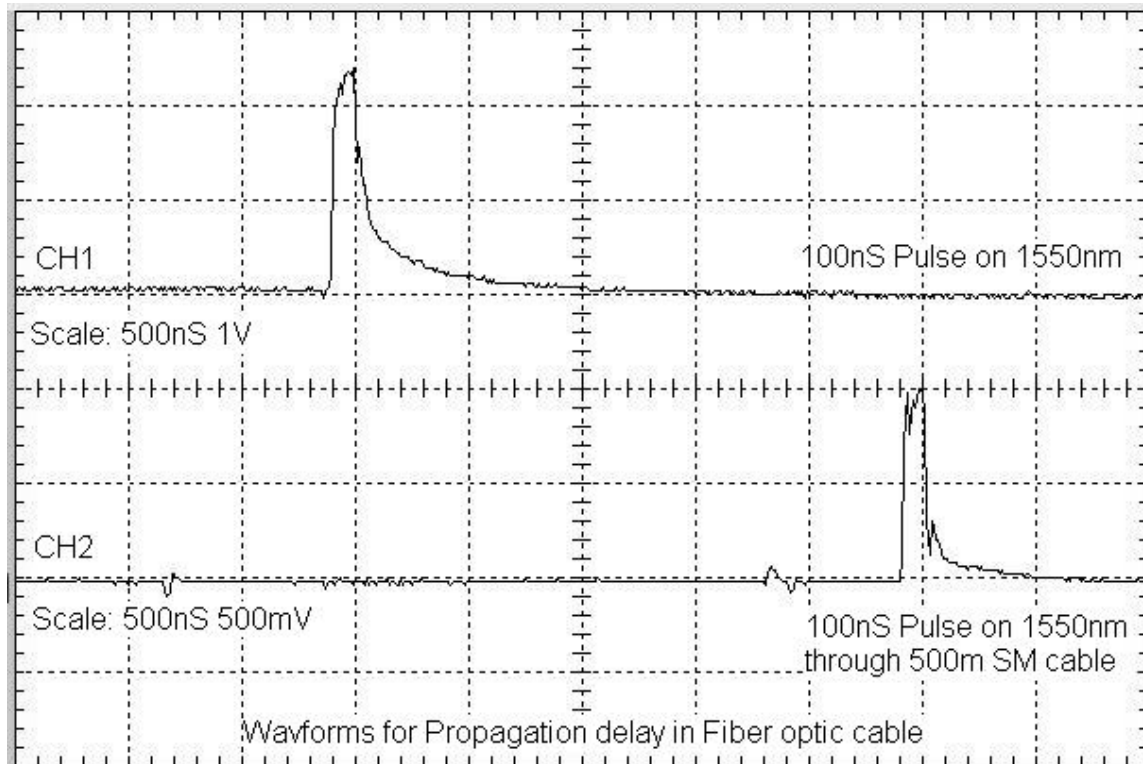
Lower NVP values will contribute to additional delay for a given length of cable, just as an increase in end-to-end cable length will cause a proportionate increase in the end-to-end delay. As with most other transmission parameters, delay values are frequency dependent.

PROCEDURE

1. Make sure that the Intensity control 1 and Intensity Control 2 pots are turned fully anticlockwise before switching on the FOL-DUAL module.
2. Keep SELECTOR 2 switch on 100nS pulse setting.
3. Apply 1 KHz TTL signal to SIGNAL I/P 2 post.
4. Connect OPTICAL O/P 2 to PORT 2 of a Coupler on FOL-PASSIVE module, using optical patch cord.
5. Connect PORT 4 of a Coupler to OPTICAL I/P 2 of a DETECTOR 2 post on FOL-DUAL, using optical patch cord.
6. Connect SIGNAL O/P 2 post of DETECTOR 2 on FOL-DUAL, to CH1 of a DSO.
7. Connect PORT 3 of a Coupler to one of the posts of a 500mtr FIBER 2 on FOL-FIBER, using optical patch cord.

8. Connect remaining post of a FIBER 2 to OPTICAL I/P 1 post of a DETECTOR 1 on FOL-DUAL, using optical patch cord.
9. Connect SIGNAL O/P 1 post of DETECTOR 1 on FOL-DUAL, to CH2 of a DSO.
10. Adjust the OFFSET POT 1 and OFFSET POT 2 to get the pulses displayed on DSO.
11. Measure the time difference $\Delta\tau$ between two pulses.
12. Calculate the fiber length, $L = V_g\Delta\tau$ where V_g is the speed of propagation of a pulse in a fiber which is approximately (c / n_1) where $c = 3 \times 10^8$ m/s is the speed of light in the vacuum and $n_1 = 1.468$ the fiber core index.

The following signals are observed on the oscilloscope:



MEASUREMENTS

Example:

$\Delta\tau$ (Propagation time delay)	= 2.5 μ S (measured on DSO)
V_g (Speed of propagation of a pulse in fiber)	$\approx (c / n_1)$
c (speed of light in the vacuum)	= 3×10^8 m/s
n_1 (fiber core index)	= 1.468
L (the fiber length)	= $V_g\Delta\tau$
	= 500mtr. (calculated)

CONCLUSION

The propagation delay $\Delta\tau$ is measured and the calculated length of fiber is verified.